Supplementary Material

Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity

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Marked-up manuscript. This file is the revised manuscript with the changes made relative to the previous version marked. Text in blue is new or rewritten; text in red belonged to the previous version and has been removed or replaced. Obsolete figures and tables of the previous version are not reproduced here; the corresponding changes are itemised in the response to the editor. Because the physics-first version is a reorganisation of the paper, most paragraphs carry a mark.

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Overview

This Supplementary Material accompanies the manuscript *"Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity"*. It contains detailed numerical formulations, complete network architecture specifications, cell- and seed-level campaign datasets for all sensitivity campaigns, grid-convergence calculations, the wall-shear estimator analysis, and reproducibility guidelines. The material is organized so that the main text can be read as a fluid-mechanics paper: every result that concerns the identification procedure, the variational landscape, the sensitivity campaigns or the numerical verification is documented here, while the main text retains only the physical formulation, the actuation modes, the dissipation comparison and its interpretation. All section numbers in this document are prefixed with "S" to distinguish them from main-text sections, and all figures are numbered S1, S2, ... independently of the main text. The tables are numbered in the same way, so that Table S1 of this document is distinct from Table 1 of the main text. Cross-references to the main manuscript use the notation (Main Sec., Main Fig. X or Main Table X).

S1. Detailed PINN Implementation

S1.1 Network Architecture Specifications

This Supplementary Material accompanies the manuscript *"Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity"*. It contains detailed numerical formulations, complete network architecture specifications, cell- and seed-level campaign datasets for all sensitivity campaigns, grid-convergence calculations, the wall-shear estimator analysis, and reproducibility guidelines. The material is organized so that the main text can be read as a fluid-mechanics paper: every result that concerns the identification procedure, the variational landscape, the sensitivity campaigns or the numerical verification is documented here, while the main text retains only the physical formulation, the actuation modes, the dissipation comparison and its interpretation. All section numbers in this document are prefixed with "S" to distinguish them from main-text sections, and all figures are numbered S1, S2, ... independently of the main text. The tables are numbered in the same way, so that Table S1 of this document is distinct from Table 1 of the main text. Cross-references to the main manuscript are written out explicitly as "Main Sec. X", "Main Fig. X" or "Main Table X".

S1 Detailed PINN Implementation

S1.1 Network Architecture Specifications

The physics-informed optimization employs two separate fully connected neural networks operating in double-precision (float64) arithmetic throughout: a *flow network* that represents the velocity and pressure fields $(u, v, p)(x, y, t; \widetilde{Re})$, and a *control network* that represents the spatio-temporal Fourier coefficients $a_{mn}(\widetilde{Re})$. The two networks share the same input normalization but have different widths and depths, reflecting the different complexity of the flow field and the low-dimensional control space.

Input normalization. The spatial coordinates $(x, y) \in [0, L_x] \times [0, 1]$ and time $t \in [0, 1]$ are fed directly to the networks. The Reynolds number is normalized to

$$\widetilde{Re} = \frac{Re - 550}{450} \in [-1, 1], \tag{1}$$

so that the training range $Re \in [100, 1000]$ maps to $[-1, 1]$, improving numerical conditioning of the gradient computations.

Flow network. Six hidden layers, each with 64 neurons, using hyperbolic tangent (tanh) activation functions. Inputs: $(x, y, t, \widetilde{Re})$, dimension 4. Outputs: (u, v, p), dimension 3. Total trainable parameters: $4 \times 64 + 5 \times (64 \times 64 + 64) + 64 \times 3 + 3 = 21\,379$.

Control network. Four hidden layers, each with 32 neurons, using tanh activation. Inputs: $(\widetilde{Re})$, dimension 1. Outputs: $\{a_{mn}\}_{m=1,\dots,N_x;\, n=0,\dots,N_t-1}$, dimension $N_x \times N_t = 30$ (baseline). Total

trainable parameters: $1 \times 32 + 4 \times (32 \times 32 + 32) + 32 \times 30 + 30 = 3\,614$.

Input normalization. The spatial coordinates $(x, y) \in [0, L_x] \times [0, 1]$ are fed directly to the networks, and so is the raw training-time variable $\tau \in [0, T_{\max}] = [0, 1]$: the training clock is a dimensionless variable of the identification problem, not the physical forcing time. The exported signal is evaluated on a different clock, $t_{\text{phys}} \in [0, T] = [0, 2]$ (Sec. S2.1); the two are related below. The Reynolds number is normalized to

$$\widetilde{Re} = \frac{Re - 550}{450} \in [-1, 1], \tag{2}$$

so that the training range $Re \in [100, 1000]$ maps to $[-1, 1]$, improving numerical conditioning of the gradient computations.

Flow network. Six hidden layers, each with 64 neurons, using hyperbolic tangent (tanh) activation functions. Inputs: $(x, y, \tau, \widetilde{Re})$, where $\tau \in [0, 1]$ is the raw nondimensional training time, dimension 4. Outputs: (u, v, p), dimension 3. Total trainable parameters: $4 \times 64 + 64 + 5 \times (64 \times 64 + 64) + 64 \times 3 + 3 = 21\,315$.

Control network. Four hidden layers, each with 32 neurons, using tanh activation. Inputs: $(\widetilde{Re})$, dimension 1. Outputs: $\{a_{mn}\}_{m=1,\ldots,N_x;\, n=0,\ldots,N_t-1}$, dimension $N_x \times N_t = 30$ (baseline). Total trainable parameters: $1 \times 32 + 32 + 3 \times (32 \times 32 + 32) + 32 \times 30 + 30 = 4\,222$.

Initialization. All network weights are initialized using the Xavier/Glorot normal distribution with zero-mean and variance $\sigma^2 = 2/(n_{\text{in}} + n_{\text{out}})$ where n_{in} and n_{out} are the fan-in and fan-out of each layer. All biases are initialized to zero.

Precision. All weight tensors, input tensors, automatic differentiation operations, and loss evaluations use 64-bit (double) precision. This choice avoids numerical cancellation in the evaluation of the PDE residuals, which involve second-order derivatives of the network outputs.

Table S1 summarizes the architecture specifications.

Table S1: Neural network architecture specifications for flow and control networks.

Network	Layers	Neurons	Activation	Inputs	Outputs
Flow	6 hidden + I/O	64	tanh	4	3 (u,v,p)
Control	4 hidden + I/O	32	tanh	1	30 (a_{mn})

S1.2 Collocation Sampling and Initialization

S1.2 Collocation Sampling and Initialization

All collocation points are resampled from scratch at the beginning of each training epoch, preventing overfitting to any fixed quadrature set.

Interior PDE collocation: $N_{\text{phys}} = 4\,000$ points sampled uniformly at random over the computational domain $\Omega \times [0, 1] = [0, L_x] \times [0, 1] \times [0, 1]$. The time coordinate fed to the network is a normalised phase $\theta = t_{\text{phys}}/T \in [0, 1]$, where $T = 2$ is the physical forcing period; the temporal basis is $\cos(2\pi n\theta)$, periodic over $[0, 1]$ by construction. At each epoch, three Reynolds-number values are selected from the set $\{100, 300, 500, 700, 900, 1000\}$; each interior point is evaluated at all three selected Re values, giving an effective PDE sample size of $3N_{\text{phys}} = 12\,000$ residual evaluations per epoch.

Interior PDE collocation: $N_{\text{phys}} = 4\,000$ points sampled uniformly at random over the computational domain $\Omega \times [0, T_{\max}] = [0, L_x] \times [0, 1] \times [0, 1]$. The time coordinate fed to the network is the raw training variable $\tau \in [0, T_{\max}] = [0, 1]$, drawn uniformly, and the temporal basis of the control is $\cos(n\pi\tau)$ over that interval; the full forcing period $T = 2$ is reached only on export (Sec. S2.1), so the training window covers one half-period of the fundamental harmonic. At each epoch, three Reynolds-number values are selected from the set $\{100, 300, 500, 700, 900, 1000\}$; each interior point is evaluated at all three selected Re values, giving an effective PDE sample size of $3N_{\text{phys}} = 12\,000$ residual evaluations per epoch.

Boundary collocation: $N_{\mathrm{BC}} = 600$ points distributed across the boundary $\partial\Omega$, split as 200 on Γ_{top} (upper lid) and 400 on Γ_{stat} (bottom and lateral walls), resampled uniformly at each epoch.

Reynolds parameter: During each epoch, 6 discrete Reynolds values (100, 300, 500, 700, 900, 1000) are used for boundary-condition enforcement and the Reynolds-separation regularization.

S1.3 Optimization Schedule and Hyperparameters

S1.3 Optimization Schedule and Hyperparameters

Training proceeds in two stages:

Stage 1 — Pre-training (300 epochs): Only the flow network weights are updated. The control coefficients a_{mn} are frozen at zero, preventing the optimizer from immediately collapsing to degenerate control solutions before a physically consistent flow representation has been established. During this stage, only the PDE residual, boundary-condition, and Reynolds-separation losses are active (the dissipation and energy objectives depend on flow accuracy to be meaningful).

Stage 2 — Joint optimization (3 000 epochs): All flow and control network weights are updated simultaneously. The full multi-objective loss (Main Eq. (13)) is active with the reference weights listed after that equation.

Stage 2 — Joint optimization (3 000 epochs): All flow and control network weights are updated simultaneously. The full multi-objective loss described in Sec. S1.4 is active with the reference weights listed after Eq. (13).

Optimizer: Adam with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon_{\mathrm{Adam}} = 10^{-8}$.

Learning rate schedule: Initial rate $\eta_0 = 8 \times 10^{-4}$ with exponential decay by factor 0.995 every 100 epochs, giving $\eta(e) = \eta_0 \times 0.995^{\lfloor e/100 \rfloor}$. After 3 000 epochs, the effective learning rate is approximately $8 \times 10^{-4} \times 0.995^{30} \approx 6.9 \times 10^{-4}$, so the learning rate has decayed by only $\sim 14\%$ over the full training. No gradient clipping is applied.

Convergence behavior: Typical behavior over the 3 000-epoch joint phase shows $\mathcal{L}_{\mathrm{PDE}}$ reaching approximately 10^{-3} by epoch 3 000, $\mathcal{L}_{\mathrm{BC}}$ below 5×10^{-4}, and $\mathcal{L}_{\mathrm{ctrl}}$ below 10^{-4} (corresponding to $|E_r - E^*| < 10^{-2}$). The total loss value at convergence lies in the range $[1.37, 1.40]$ for the $(Re, E^*) = (500, 0.25)$ condition used in the seed study.

Seed sensitivity: The early training stochasticity during Stage 1 determines which outcome cluster is entered during Stage 2. Once Stage 2 begins, the optimizer rarely switches clusters, explaining the reproducible classification of each realization. No early stopping is applied; all realizations run for the full 3 000 epochs regardless of convergence.

S1.4 Multi-Objective Loss Formulation

The complete loss is (Main Eq. (13)):

$$\mathcal{L} = \lambda_{\mathrm{PDE}}\mathcal{L}_{\mathrm{PDE}} + \lambda_{\mathrm{BC}}\mathcal{L}_{\mathrm{BC}} + \lambda_{\mathrm{diss}}\mathcal{L}_{\mathrm{diss}} + \lambda_{\mathrm{ctrl}}\mathcal{L}_{\mathrm{ctrl}} + \lambda_{\mathrm{var}}\mathcal{L}_{\mathrm{var}} + \lambda_{Re}\mathcal{L}_{Re}, \tag{3}$$

with baseline weights $\lambda_{\mathrm{PDE}} = 1$, $\lambda_{\mathrm{BC}} = 15$, $\lambda_{\mathrm{diss}} = 0.05$, $\lambda_{\mathrm{ctrl}} = 200$, $\lambda_{\mathrm{var}} = 1$, $\lambda_{Re} = 2$. The individual components are:

$$\mathcal{L}_{\mathrm{PDE}} = \left\langle R_u^2 + R_v^2 + R_c^2 \right\rangle_{\mathcal{X}_{\mathrm{int}}}, \tag{4}$$

$$\mathcal{L}_{\mathrm{BC}} = \left\langle u^2 + v^2 \right\rangle_{\Gamma_{\mathrm{stat}}} + \left\langle (u - U_{\mathrm{lid}})^2 + v^2 \right\rangle_{\Gamma_{\mathrm{top}}}, \tag{5}$$

$$\mathcal{L}_{\mathrm{diss}} = \int_\Omega \frac{1}{Re}\left[u_x^2 + v_y^2 + \tfrac{1}{2}(u_y + v_x)^2\right] d\Omega = \int_\Omega \nu\, S{:}S \, d\Omega, \tag{6}$$

$$\mathcal{L}_{\mathrm{ctrl}} = (E_r - E^*)^2, \tag{7}$$

$$\mathcal{L}_{\mathrm{var}} = -\log\left[\mathrm{Var}(U_{\mathrm{lid}}) + 10^{-5}\right], \tag{8}$$

$$\mathcal{L}_{Re} = \max\left[0,\ \delta - \frac{1}{N_{\mathrm{probe}}} \sum_{i=1}^{N_{\mathrm{probe}}} \left(U_{\mathrm{lid}}(x_i, t_0, Re_{\min}) - U_{\mathrm{lid}}(x_i, t_0, Re_{\max})\right)^2\right], \tag{9}$$

with $N_{\mathrm{probe}} = 20$ probe locations distributed uniformly in $x \in [0, L_x]$, $t_0 = 0.5$, $\delta = 0.01$, $Re_{\min} = 100$, $Re_{\max} = 1000$.

No initial-condition term is included by design. The temporal Fourier expansion with basis $\cos(2n\pi\theta)$ defines a periodic space–time boundary-value problem on $\theta \in [0, 1]$, so training targets a statistically stationary periodic state rather than the transient from a prescribed initial field (Main Sec. 2.1). Periodicity of the flow field is enforced by the basis and verified a posteriori in Sec. S8.5.

Weight rationale. The weight $\lambda_{\mathrm{BC}} = 15$ strongly enforces boundary conditions relative to the interior residual, preventing poorly conditioned flow states where boundary violations dominate the solution. $\lambda_{\mathrm{ctrl}} = 200$ is large to enforce the energy constraint tightly ($|E_r - E^*|/E^* < 5\%$ at convergence for all tested conditions), ensuring that all realizations are compared at effectively the same actuation energy. The small value $\lambda_{\mathrm{diss}} = 0.05$ prevents the dissipation objective from overwhelming the PDE constraints during early training, which can cause the optimizer to ignore physical accuracy in favor of reducing dissipation. The regularizer 10^{-5} in $\mathcal{L}_{\mathrm{var}}$ prevents numerical issues when $\mathrm{Var}(U_{\mathrm{lid}}) = 0$ at the beginning of training before the control network has been activated.

S2. Control-Space and Modal-Analysis Details

S2.1 Spatio-Temporal Fourier Basis

The boundary actuation is expanded as in Main Sec. 3.2:

$$U_{\mathrm{lid}}(x, \theta, Re) = \sum_{m=1}^{N_x} \sum_{n=0}^{N_t-1} a_{mn}(Re) \sin\left(\frac{m\pi x}{L_x}\right) \cos(2n\pi\theta), \tag{10}$$

with baseline truncation $N_x = 6$, $N_t = 5$, giving 30 nominal coefficients. The sine basis $\sin(m\pi x/L_x)$ is chosen because it vanishes at $x = 0$ and $x = L_x$ for all $m \geq 1$, automatically enforcing corner compatibility without any additional constraint. The cosine basis $\cos(2n\pi\theta)$ has the property that $\cos(0) = 1$, so the $n = 0$ terms represent purely stationary (time-independent) actuation, and every term is periodic with unit period in θ. The physical time is $t_{\mathrm{phys}} = T\theta$ with $T = 2$; the time derivative in the Navier–Stokes residual therefore carries the chain-rule factor $\partial/\partial t_{\mathrm{phys}} = (1/T)\,\partial/\partial\theta$. All mode indices in this manuscript use the physical (one-based) convention: the second spatial harmonic corresponds to $m = 2$, not $m = 1$.

S2.2 Modal Energy Normalization and Quadrature

The time-average of the squared basis functions satisfies $\langle\cos^2(0)\rangle_t = 1$ and $\langle\cos^2(n\pi t)\rangle_t = \frac{1}{2}$ for $n \geq 1$. The space-average of the squared sine basis functions satisfies $\langle\sin^2(m\pi s)\rangle_s = \frac{1}{2}$ for all $m \geq 1$, where $s = x/L_x \in [0, 1]$. Therefore, the contribution of coefficient a_{mn} to the realized energy $E_r = \langle U_{\mathrm{lid}}^2\rangle_{s,t}$ is:

$$E_{mn} = \langle a_{mn}^2 \sin^2(m\pi s)\cos^2(n\pi t)\rangle_{s,t} = \begin{cases} \frac{1}{2}a_{m0}^2 & n = 0, \\ \frac{1}{4}a_{mn}^2 & n \geq 1. \end{cases} \tag{11}$$

These are the modal energies $\mathcal{E}_{mn}$ used in Main Sec. 3.2. The realized actuation energy is therefore

$$E_r = \sum_{m=1}^{N_x} \left(\tfrac{1}{2}a_{m0}^2 + \sum_{n=1}^{N_t-1} \tfrac{1}{4}a_{mn}^2 \right). \tag{12}$$

S2.3 Branch Classification Thresholds

The branch clusters are defined by the following explicit rule applied in order to the normalized modal fractions $f_1 \equiv f_{10}$ (stationary first spatial harmonic), $f_2 \equiv f_{20}$ and $f_t = \sum_{m,n\geq 1} f_{mn}$:

- B_{first}: $f_1 > 0.5$ (stationary first harmonic carries an absolute majority of the actuation energy).
- B_1: $f_2 > 2f_t$ and $f_2 > 0.5$ (stationary second mode carries at least twice the energy of all temporal modes, and more than half of the total).
- B_2: $f_t > 2f_2$ and $f_t > 0.5$ (temporal family dominant).
- B_3: none of the above (residual class).

This is a dominance rule rather than an absolute cut-off on f_2 or f_t: a realization is assigned to the class that carries the control energy, and the rule reproduces the three well-separated groups visible

in the (f_2, f_t) observable space of the identification campaign. The parameters $k = 2$ (dominance factor) and $\phi = 0.5$ (floor) were fixed a priori and not tuned to the data.

The B_{first} test was added after a full-spectrum audit of all 185 realizations and of the λ_{var} campaign, because the earlier three-class rule assigned a first-harmonic-dominated control to B_3 whenever it satisfied neither $f_2 > 0.5$ nor $f_t > 0.5$. The audit shows that this defect never materializes in the broad campaign: $f_1 \leq 0.122$ over all 185 runs ($f_1 \leq 0.096$ within B_3), so B_{first} is empty there and all 185 labels, frequencies and confidence intervals are unchanged (the class audit of Sec. S2.3). It does materialize in the ablation campaign, where removing $\mathcal{L}_{\text{var}}$ produces $f_1 = 0.87$–0.99 (Table ?). The audit also resolves what B_3 contains: 15 of its 26 members are dominated by the temporal family, 8 by the stationary second harmonic and 3 by stationary harmonics of order $m \geq 3$; none by the first harmonic (sub-structure analysis of Sec. S2.3).

S1.4 Multi-Objective Loss Formulation

The complete loss is

$$\mathcal{L} = \lambda_{\text{PDE}}\mathcal{L}_{\text{PDE}} + \lambda_{\text{BC}}\mathcal{L}_{\text{BC}} + \lambda_{\text{diss}}\mathcal{L}_{\text{diss}} + \lambda_{\text{ctrl}}\mathcal{L}_{\text{ctrl}} + \lambda_{\text{var}}\mathcal{L}_{\text{var}} + \lambda_{Re}\mathcal{L}_{Re}, \tag{13}$$

with baseline weights $\lambda_{\text{PDE}} = 1$, $\lambda_{\text{BC}} = 15$, $\lambda_{\text{diss}} = 0.05$, $\lambda_{\text{ctrl}} = 200$, $\lambda_{\text{var}} = 1$, $\lambda_{Re} = 2$. The individual components are:

$$\mathcal{L}_{\text{PDE}} = \left\langle R_u^2 + R_v^2 + R_c^2 \right\rangle_{\mathcal{X}_{\text{int}}}, \tag{14}$$

$$\mathcal{L}_{\text{BC}} = \left\langle u^2 + v^2 \right\rangle_{\Gamma_{\text{stat}}} + \left\langle (u - U_{\text{lid}})^2 + v^2 \right\rangle_{\Gamma_{\text{top}}}, \tag{15}$$

$$\mathcal{L}_{\text{diss}} = \int_\Omega \frac{1}{Re}\left[u_x^2 + v_y^2 + \tfrac{1}{2}(u_y + v_x)^2\right] d\Omega = \int_\Omega \nu\, S{:}S\, d\Omega, \tag{16}$$

$$\mathcal{L}_{\text{ctrl}} = (E_r - E^*)^2, \tag{17}$$

$$\mathcal{L}_{\text{var}} = -\log\left[\text{Var}(U_{\text{lid}}) + 10^{-5}\right], \tag{18}$$

$$\mathcal{L}_{Re} = \max\left[0,\ \delta - \frac{1}{N_{\text{probe}}}\sum_{i=1}^{N_{\text{probe}}} \left(U_{\text{lid}}(x_i, t_0, Re_{\min}) - U_{\text{lid}}(x_i, t_0, Re_{\max})\right)^2\right], \tag{19}$$

with $N_{\text{probe}} = 20$ probe locations distributed uniformly in $x \in [0, L_x]$, $t_0 = 0.5$, $\delta = 0.01$, $Re_{\min} = 100$, $Re_{\max} = 1000$.

The dissipation objective $\mathcal{L}_{\text{diss}}$ is one half of the physical viscous-dissipation rate defined in Main Eq. (6):

$$\mathcal{L}_{\text{diss}} = \frac{1}{2}D. \tag{20}$$

The constant factor $1/2$ is retained in the training weights; it does not affect the location of the minimum. This constant normalization is retained in the reported training weights; the relative weighting of the loss terms is therefore based on $\mathcal{L}_{\text{diss}}$, not on D directly.

No initial-condition term is included by design. The identification solves a space–time boundary-value problem in the training variables $(x, y, \tau) \in [0, L_x] \times [0, 1] \times [0, 1]$, with the control expanded on the basis $\cos(n\pi\tau)$, so training targets a stationary response of that problem rather than the transient from a prescribed initial field (Main Sec. 2.1). Time derivatives in the Navier–Stokes residual are taken directly with respect to τ, with no chain-rule factor: the residual is written in the training clock, and $T_{\max} = 1$ is its time unit.

Three distinct properties must not be conflated:

1. *The coefficient network.* The control network outputs the amplitudes $a_{mn}(Re)$, which are independent of any time variable; they are the object identified by the training.

2. *The exported wall actuation* $U_{\text{lid}}(x, t_{\text{phys}})$: the same Fourier series is evaluated on the physical clock over the complete period $0 \leq t_{\text{phys}} \leq T = 2$, i.e. on the basis $\cos(n\pi t_{\text{phys}}) = \cos(2n\pi\theta)$

with $\theta = t_{\text{phys}}/T$ (Sec. S2.1). Because T is an integer multiple of the period $2/n$ of every harmonic, this signal is exactly periodic over $[0, T]$; the training window $[0, 1]$ covers one half-period of the fundamental, which is a restriction on the training clock only, not on the exported signal.

3. *The PINN flow field* (u, v, p): the flow network receives $(x, y, \tau, \widetilde{Re})$ as inputs and uses tanh activations, which do not enforce periodicity in τ, and no periodic endpoint condition is imposed on it. It is used only during training to satisfy the Navier–Stokes equations; it is not transferred to the LBM solver. The actual flow field used for dissipation calculations is computed by the MRT-LBM solver driven by the periodic $U_{\text{lid}}(x, t_{\text{phys}})$ exported from the PINN control network. The periodicity of this LBM-computed flow field is verified a posteriori in Sec. S4.5 by comparing the phase averages of the first and second halves of the sampling window, with discrepancies below 1.0×10^{-3} in relative field norm for all three branches.

Weight rationale. The weight $\lambda_{\text{BC}} = 15$ strongly enforces boundary conditions relative to the interior residual, preventing poorly conditioned flow states where boundary violations dominate the solution. $\lambda_{\text{ctrl}} = 200$ is large to enforce the energy constraint tightly ($|E_r - E^*|/E^* < 5\%$ at convergence for the reference configurations; deviations up to 11% occur in some sensitivity tests with modified capacity or truncation, as documented in Tables S7–S8), ensuring that all realizations are compared at effectively the same actuation energy. The small value $\lambda_{\text{diss}} = 0.05$ prevents the dissipation objective from overwhelming the PDE constraints during early training, which can cause the optimizer to ignore physical accuracy in favor of reducing dissipation. The regularizer 10^{-5} in $\mathcal{L}_{\text{var}}$ prevents numerical issues when $\text{Var}(U_{\text{lid}}) = 0$ at the beginning of training before the control network has been activated.

S2 Control-Space and Modal-Analysis Details

S2.1 Spatio-Temporal Fourier Basis

The boundary actuation is expanded as in Main Sec. 3.2, on the export clock t_{phys} (equivalently on the normalized phase $\theta = t_{\text{phys}}/T \in [0, 1]$):

$$U_{\text{lid}}(x, t_{\text{phys}}, Re) = \sum_{m=1}^{N_x} \sum_{n=0}^{N_t-1} a_{mn}(Re) \sin\left(\frac{m\pi x}{L_x}\right) \cos\left(\frac{2n\pi t_{\text{phys}}}{T}\right), \qquad T = 2. \tag{21}$$

The coefficients a_{mn} are those identified in the training clock $\tau \in [0, 1]$ of Sec. S1.2, where the same series is written $\cos(n\pi\tau)$; export re-evaluates that series over the complete period $0 \le t_{\text{phys}} \le T = 2$, so that $\cos(2n\pi t_{\text{phys}}/T) = \cos(n\pi t_{\text{phys}})$. The baseline truncation is $N_x = 6$, $N_t = 5$, giving 30 nominal coefficients. The sine basis $\sin(m\pi x/L_x)$ is chosen because it vanishes at $x = 0$ and $x = L_x$ for all $m \ge 1$, automatically enforcing corner compatibility without any additional constraint. The cosine basis $\cos(2n\pi t_{\text{phys}}/T)$ has the property that $\cos(0) = 1$, so the $n = 0$ terms represent purely stationary (time-independent) actuation, and every term is periodic with period T in t_{phys}: the fundamental oscillates at $\omega_1 = 2\pi/T = \pi$. The lattice is driven by this exported series, not by the PINN flow field, and the training residual differentiates with respect to τ alone, so no chain-rule factor $1/T$ enters the identification (Sec. S1.4). All mode indices in this manuscript use the physical (one-based) convention: the second spatial harmonic corresponds to $m = 2$, not $m = 1$.

S2.2 Modal Energy Normalization and Quadrature

The phase average of the squared temporal basis functions satisfies $\langle \cos^2(0) \rangle_\theta = 1$ and $\langle \cos^2(2\pi n\theta) \rangle_\theta = \frac{1}{2}$ for $n \ge 1$. The same values hold for $\cos(n\pi\tau)$ averaged over the training window $\tau \in [0, 1]$, because $\int_0^1 \cos^2(n\pi\tau)\, d\tau = \int_0^1 \cos^2(2n\pi\theta)\, d\theta = \frac{1}{2}$ and the cross terms vanish on both windows: the energy constraint applied during identification is therefore the cycle mean of the exported signal, even though the flow field itself is trained on the shorter window (Sec. S1.4).

The space-average of the squared sine basis functions satisfies $\langle \sin^2(m\pi s)\rangle_s = \frac{1}{2}$ for all $m \geq 1$, where $s = x/L_x \in [0,1]$. Therefore, the contribution of coefficient a_{mn} to the realized energy $E_r = \langle U_{\text{lid}}^2\rangle_{s,\theta}$ is:

$$E_{mn} = \langle a_{mn}^2 \sin^2(m\pi s)\cos^2(2\pi n\theta)\rangle_{s,\theta} = \begin{cases} \frac{1}{2}a_{m0}^2 & n = 0, \\ \frac{1}{4}a_{mn}^2 & n \geq 1. \end{cases} \tag{22}$$

These are the modal energies $\mathcal{E}_{mn}$ used in Main Sec. 3.2. The realized actuation energy is therefore

$$E_r = \sum_{m=1}^{N_x} \left(\tfrac{1}{2}a_{m0}^2 + \sum_{n=1}^{N_t-1} \tfrac{1}{4}a_{mn}^2 \right). \tag{23}$$

S2.3 Branch Classification Thresholds

The branch clusters are defined by the following explicit rule applied in order to the normalized modal fractions $f_1 \equiv f_{10}$ (stationary first spatial harmonic), $f_2 \equiv f_{20}$ and $f_t = \sum_{m=1}^{N_x}\sum_{n=1}^{N_t-1} f_{mn}$:

- B_{first}: $f_1 > 0.5$ (stationary first harmonic carries an absolute majority of the actuation energy).
- B_1: $f_2 > 2f_t$ and $f_2 > 0.5$ (stationary second mode carries at least twice the energy of all temporal modes, and more than half of the total).
- B_2: $f_t > 2f_2$ and $f_t > 0.5$ (temporal family dominant).
- B_3: none of the above (residual class).

This is a dominance rule rather than an absolute cut-off on f_2 or f_t: a realization is assigned to the class that carries the control energy, and the rule reproduces the three well-separated groups visible in the (f_2, f_t) observable space of the identification campaign. The parameters $k = 2$ (dominance factor) and $\phi = 0.5$ (floor) were fixed a priori and not tuned to the data.

The B_{first} test ($f_1 > 0.5$) identifies controls where the first spatial harmonic carries an absolute majority of the actuation energy. This class is empty in the broad campaign: $f_1 \leq 0.122$ over all 185 runs ($f_1 \leq 0.096$ within B_3), so all 185 labels, frequencies and confidence intervals are unchanged. The class does appear in the ablation campaign when $\mathcal{L}_{\text{var}}$ is removed, producing $f_1 = 0.87$–0.99 (Table S6). The B_3 class contains 15 members dominated by the temporal family, 8 by the stationary second harmonic and 3 by stationary harmonics of order $m \geq 3$; none by the first harmonic.

Table S2 reports the sensitivity of the pooled frequencies to both k and ϕ: varying k between 1.5 and 3, or ϕ between 0.4 and 0.6, moves $\widehat{P}(B_1)$ and $\widehat{P}(B_2)$ by less than three percentage points and changes at most 12 of the 185 realizations. The frequency of the residual class B_3 is more sensitive (9.2%–20.5%) because it is defined as the complement of the other conditions and is the smallest class; its share should therefore be read as approximate, whereas the B_1/B_2 balance used in this study is robust. Lowering the B_{first} threshold from 0.5 to 0.2 or even 0.13 would still leave the class empty in this campaign, so the pooled frequencies are insensitive to that choice. Only 6 realizations lie within 0.05 of a dominance boundary, and the smallest distance to a boundary over the whole campaign is 0.008.

S3. Identification Campaigns: Summary

The broad (Re, E^*) campaign (185 realizations), the variance-regularization sweep (90 realizations), the ten-seed multiplicity study, the loss-term ablation, the network capacity and control-truncation studies, the aspect-ratio dependence, and the Fourier versus modified-Chebyshev basis comparison are documented in full in Ref. and in the data repository. This section collects the results that are directly needed to identify the three branch representatives used in the main text.

S3.1 Branch classification and the B_{first} class

The branch clusters are defined by the dominance rule of Sec. S2.3 applied to the normalized modal fractions f_1, f_2 and f_t. The B_{first} test ($f_1 > 0.5$) was added after a full-spectrum audit of all 185 realizations and of the λ_{var} campaign. It is empty in the broad campaign ($f_1 \leq 0.122$ everywhere) but materializes in the ablation campaign when $\mathcal{L}_{\text{var}}$ is removed (Table ?). The pooled

Table S2: Sensitivity of the pooled branch frequencies to the parameters of the dominance rule (B_1: $f_2 > k f_t$ and $f_2 > \phi$; B_2: $f_t > k f_2$ and $f_t > \phi$; B_3: otherwise). The reference rule is $k = 2$, $\phi = 0.5$; one parameter is varied at a time. The last column counts the realizations whose label changes with respect to the reference.

Parameter	Value	B_1 (%)	B_2 (%)	B_3 (%)	Reclassified (of 185)
dominance factor k	1.50	43.2	47.6	9.2	9
dominance factor k	1.75	42.7	45.4	11.9	4
dominance factor k	2.00 (reference)	42.2	43.8	14.1	0
dominance factor k	2.50	40.5	41.6	17.8	7
dominance factor k	3.00	39.5	40.0	20.5	12
floor ϕ	0.40	42.2	44.9	13.0	2
floor ϕ	0.50 (reference)	42.2	43.8	14.1	0
floor ϕ	0.60	41.1	40.5	18.4	8

Table S3: Spatial sine decomposition of the imposed lid series of the three branch controls, harmonic by harmonic. $A_n(x)$ is the amplitude of temporal harmonic n of $U_{\mathrm{lid}}(x,t)$ ($n = 0$ is the time mean); a_{mn} is its coefficient in the basis $\sin(m\pi x)$ and E_n its share of the actuation energy, with the convention $E_0 = \int_0^1 A_0^2\,dx$ and $E_n = \frac{1}{2}\int_0^1 A_n^2\,dx$ for $n \geq 1$ (Sec. S2.2). The shares are computed from the series as imposed in the captures: the archived B_1 Fourier series has native norm 0.2534 and is rescaled by 0.9933 to the common $E^* = 0.25$, the B_2 and B_3 series are stored already at that norm. The last column is the fraction of lid nodes whose velocity changes sign at least once during the period, counted on the interior nodes: B_1 reverses only at the midline node, where its stationary two-lobe profile has its node, whereas the reversing band of B_3 extends over $0.28 < x < 0.64$. Computed with `scripts/lid_spatial_decomposition.py` from the lid series archived in `data/lid_series/`.

Control	E_0	a_{20}	E_1/E_{tot}	a_{11}	share of $m = 1$ in A_1	reversing span
B_1 (full series)	0.987	+0.679	0.012	0.068	0.397	0.004
B_2 (seed 5)	0.002	−0.001	0.977	0.981	0.985	1.000
B_3 (seed 0)	0.552	−0.510	0.433	0.627	0.909	0.362

frequencies $\widehat{P}(B_1) = 42.2\%$ and $\widehat{P}(B_2) = 43.8\%$ are stable to within three percentage points under variation of the dominance factor $k \in [1.5, 3]$ and floor $\phi \in [0.4, 0.6]$; the residual class B_3 is more sensitive because it is defined as the complement.

S3.2 Ten-seed experiment and loss-term ablation at $Re = 500$, $E^* = 0.25$

Spatial structure of the three branch controls

The dominance rule above counts energy in families, not in wall motions, so the physical content of each class is stated here explicitly. Table S3 decomposes each imposed lid series first in time and then, harmonic by harmonic, in the spatial sine basis. The stationary part of B_1 carries 98.7% of its actuation energy and 93.4% of that part sits in the second spatial mode ($a_{20} = 0.68$), the antisymmetric two-lobe lid with a node at the midline. Its wall velocity keeps its sign over the whole span except at that node, where the stationary profile vanishes and the residual 1.3% oscillation sets the sign: 0.4% of the lid reverses during the period. B_2 is the opposite case: 97.7% of the energy is in the first temporal harmonic and 98.5% of that harmonic is the single-lobe first spatial mode ($a_{11} = 0.98$), so the whole wall reverses direction twice per period and its time mean is nearly zero. B_3 superposes the two structures, with 55% of the energy in a stationary second-harmonic lid (94% of that part in $m = 2$) and 43% in an oscillating single-lobe lid (91% in $m = 1$); the two cancel over the central band $0.28 < x < 0.64$, which is the 36% of the span that reverses. The three classes are therefore, physically, a quasi-steady two-lobe lid, a lid-oscillatory single-lobe lid and a mixed lid, which is how the main text refers to them.

S3 Identification Campaigns: Scope and Reuse

Two of the campaigns summarised here were already published: the broad (Re, E^*) campaign (185 realizations) and the variance-regularization sweep (90 realizations) are reported in Ref. [1]. In the present work they are *re-analysed*, not re-run: no new training is involved, and the numbers quoted below are relabelings or re-combinations of the published quantities, restricted to the classification question raised in the main text. The ten-seed multiplicity study, the loss-term ablation, the network capacity and control-truncation studies, the aspect-ratio dependence, and the Fourier versus modified-Chebyshev basis comparison are documented in Ref. [1] and reproduced here for completeness so that the branch selection is auditable without leaving this document. The new contributions of the present work are: (i) the classification of the three branch representatives B_1, B_2, B_3 used in the main text; (ii) the independent MRT-LBM verification of their dissipation budgets; (iii) the physical interpretation of the cycle-mean dissipation in terms of Stokes boundary layers and steady-streaming mechanisms; and (iv) the systematic comparison of $\overline{D}$ against $D(\overline{\mathbf{u}})$. This section collects the results that are directly needed to identify the three branch representatives used in the main text; the tables are reproduced here so that the branch selection is auditable without leaving this document.

S3.1 Branch classification and the B_{first} class

The branch clusters are defined by the dominance rule of Sec. S2.3 applied to the normalized modal fractions f_1, f_2 and f_t. The B_{first} test ($f_1 > 0.5$) is empty in the broad campaign ($f_1 \leq 0.122$ everywhere) but appears in the ablation campaign when $\mathcal{L}_{\text{var}}$ is removed (Table S6). The pooled frequencies $\widehat{P}(B_1) = 42.2\%$ and $\widehat{P}(B_2) = 43.8\%$ are stable to within three percentage points under variation of the dominance factor $k \in [1.5, 3]$ and floor $\phi \in [0.4, 0.6]$; the residual class B_3 is more sensitive because it is defined as the complement.

S3.2 Ten-seed experiment and loss-term ablation at $Re = 500$, $E^* = 0.25$

Table S4 reports the seed-level metrics for the ten-seed multiplicity study. The tight loss range $[1.371, 1.394]$ across qualitatively distinct modal structures confirms that the loss landscape does not order the branch types. Table S5 summarizes the ablation configurations: removing $\mathcal{L}_{\text{var}}$ shifts the outcome to B_{first}, while removing either of the other two terms leaves the branch unchanged.

Table S4: Seed-level metrics for the ten-seed experiment at $Re = 500$, $E^* = 0.25$.

Seed	E_r	f_2 (%)	f_t (%)	Total loss	Classification
0	0.240	53.2	43.8	1.377	Intermediate (B_3)
1	0.267	91.1	1.3	1.377	Second-mode (B_1)
2	0.257	89.2	5.0	1.374	Second-mode (B_1)
3	0.268	2.1	97.8	1.372	Temporal (B_2)
4	0.250	52.2	45.0	1.394	Intermediate (B_3)
5	0.245	0.4	99.5	1.373	Temporal (B_2)
6	0.265	86.3	1.0	1.376	Second-mode (B_1)
7	0.250	91.3	1.6	1.374	Second-mode (B_1)
8	0.250	1.3	98.4	1.372	Temporal (B_2)
9	0.267	0.1	99.6	1.371	Temporal (B_2)

S3.3 Network capacity, truncation, aspect ratio and basis

The network capacity (Table ?), control-space truncation (Table ?), aspect-ratio (Table ?) and modified-Chebyshev basis (Sec. S7.2–S7.4) studies confirm that the second-harmonic concentration is robust to the numerical parameters and is a physical property of the flow response. The Chebyshev controls, which have $f_t > 93\%$, produce the most dissipative cycle-mean states of all configurations examined, confirming that the representation choice determines the actual physical flow state.

Table S5: Complete loss-term ablation at $Re = 500$, $E^* = 0.25$.

Configuration	E_r	f_2 (%)	f_t (%)	Interpretation
Baseline	0.2534	92.2	1.3	Second-mode dominated
No $\mathcal{L}_{\text{var}}$	0.2582	0.2	5.1	Shift to B_{first}
No $\mathcal{L}_{Re}$	0.2534	92.2	1.3	Essentially unchanged
No $\mathcal{L}_{\text{diss}}$	0.2533	92.3	1.2	Essentially unchanged
No $\mathcal{L}_{\text{var}}, \mathcal{L}_{Re}$	0.2582	0.2	5.1	Same first-harmonic shift
No $\mathcal{L}_{\text{var}}, \mathcal{L}_{\text{diss}}$	0.2578	0.3	5.0	Same first-harmonic shift
$\lambda_{\text{var}}/2$	0.2546	89.3	1.5	Reduced concentration
$2\lambda_{\text{var}}$	0.2598	92.8	1.0	Slightly increased

Table S6: First-harmonic content of the $\lambda_{\text{var}} = 0$ controls (five seeds per operating point, archived in `results/02_lambda_var`). Every realization is dominated by the first spatial harmonic and is reclassified as B_{first}; the second-harmonic fraction never exceeds 2.5%.

Operating point	E_r target	f_1 (%)	f_2 (%)	f_t (%)	B_{first} seeds
$Re = 1000$, $E^* = 1$	1	86.8–97.2	≤ 1.02	≤ 12.1	5/5
$Re = 500$, $E^* = 0.25$	0.25	89.9–93.6	≤ 0.31	≤ 5.4	5/5
$Re = 700$, $E^* = 2$	2	93.1–98.6	≤ 2.41	≤ 4.5	5/5

S3.3 Network capacity, truncation, aspect ratio and basis

The network capacity (Table S7), control-space truncation (Table S8), aspect-ratio (Table S9) and modified-Chebyshev basis (Ref. [1]) studies confirm that the second-harmonic concentration is robust to variations of network capacity and truncation for the square cavity with Fourier parametrization. However, this concentration is not universal: it depends on the cavity geometry (at $AR = 0.5$, $f_t = 26.7\%$ compared to 1.3% at $AR = 1$) and on the control parametrization (the Chebyshev basis produces strongly temporal controls). The second-harmonic dominance is therefore a property of the square cavity with Fourier actuation, not a general physical law independent of geometry and representation; a control with $f_t > 93\%$, such as the Chebyshev ones of Ref. [1], satisfies the B_2 dominance condition of Sec. S2.2 by construction, and no LBM cycle-mean response was computed for such a control in the present work.

Table S7: Network capacity sensitivity at $Re = 500$, $E^* = 0.25$. The parameter counts include all weights and biases. The flow network parameters are given separately from the control network (4,222 parameters in all configurations); the total is the sum of both.

Configuration	Layers	Neurons	Flow parameters	Total parameters	E_r	f_2 (%)	f_t (%)
Small	4	32	3 427	7 649	0.2541	91.8	1.2
Reference	6	64	21 315	25 537	0.2534	92.2	1.3
Large	8	128	116 611	120 833	0.2518	89.5	2.1

S8. Independent MRT-LBM Verification and Grid Uncertainty

S8.1 MRT-LBM Formulation

The independent lattice-Boltzmann solver uses the D2Q9 model with 9 velocity directions $\mathbf{c}_i$ ($i = 0, \ldots, 8$) defined as $\mathbf{c}_0 = (0, 0)$, $\mathbf{c}_{1-4} = (\pm 1, 0), (0, \pm 1)$, and $\mathbf{c}_{5-8} = (\pm 1, \pm 1)$ (lattice units). The MRT collision step is

$$\mathbf{m}^* = \mathbf{m} - \mathbf{S}(\mathbf{m} - \mathbf{m}^{\text{eq}}), \tag{24}$$

where the moment vector $\mathbf{m} = \mathbf{M}\mathbf{f}$, $\mathbf{M}$ is the 9×9 transformation matrix from population to moment space, and $\mathbf{S} = \text{diag}(s_0, s_1, \ldots, s_8)$ with $s_\nu = s_\xi = 2/(1 + 6\nu_{\text{latt}})$ for the shear-stress relaxation rates, where $\nu_{\text{latt}} = U_{\text{ref}} L/(Re \cdot N)$ is the lattice viscosity (with N the grid size per cavity side and U_{ref} the reference lattice velocity). All other relaxation rates are set to unity ($s_i = 1$ for non-viscous moments). The post-collision populations are streamed: $f_i(\mathbf{x} + \mathbf{c}_i, t + 1) = f_i^*(\mathbf{x}, t)$.

Table S8: Complete control-truncation sensitivity results at $Re = 500$, $E^* = 0.25$.

(N_x, N_t)	Modes	A_2	E_r	f_2 (%)	f_t (%)	Class
(4,3)	12	0.671	0.2541	87.6	2.1	B_1
(6,1)	6	0.678	0.2533	89.2	0.0	B_1
(6,3)	18	0.678	0.2778	89.3	1.8	B_1
(6,5)	30	0.684	0.2533	92.3	1.3	B_1 (reference)
(8,5)	40	0.669	0.2475	87.8	2.0	B_1
(8,9)	72	0.668	0.2775	87.3	2.4	B_1

Table S9: Aspect-ratio sensitivity at $Re = 500$, $E^* = 0.25$.

AR	L_x	A_2	E_r	f_2 (%)	f_t (%)
0.5	0.5	0.6034	0.2603	69.9	26.7
1.0	1.0	0.6835	0.2534	92.2	1.3
2.0	2.0	0.6770	0.2514	91.2	1.6

Lid boundary conditions are imposed using the Zou–He scheme: at the upper wall ($y = N$), the normal populations are updated to enforce the specified lid velocity $U_{\text{lid}}(x)$ while conserving local momentum. Bottom and lateral walls use standard half-way bounce-back boundary conditions.

S8.2 Ghia et al. Benchmark Validation

The MRT-LBM solver was validated against the reference benchmark data of Ghia et al. for the uncontrolled square cavity at $Re = 100$ and $Re = 1000$ using a 256×256 lattice. Main Sec. 4.2 reports the L_2 and L_∞ errors in the centerline velocity profiles. The errors are within the expected accuracy of the LBM at this resolution, confirming that the solver correctly reproduces the known benchmark solutions before being applied to the controlled-flow study.

S8.3 Control Transfer Protocol

The PINN-identified controls are transferred to the LBM solver by evaluating the control network output at the desired Re and discretizing the resulting $U_{\text{lid}}(x, t = 0, Re)$ profile onto the lattice lid nodes. For the stationary $A_2 \sin(2\pi x)$ and $\sin(\pi x)$ configurations, the lid velocity is constant in time and is applied identically at every LBM timestep. For the PINN mean control, the mean control profile is obtained by averaging the PINN control output $\{a_{mn}\}$ over the baseline training Re values and evaluating $U_{\text{lid}}(x, t = 0)$, yielding a stationary profile that is then applied as a steady lid condition in the LBM. No retraining or adaptation of the PINN is performed; the transfer is a one-time evaluation of the neural network.

The stationary approximation implicit in this protocol is quantified in Sec. S8.5. The B_1 branch carries a temporal energy fraction of about 7.4%, so the time-mean profile is not exactly the control that the PINN produces. When the full time-dependent Fourier control is transferred instead, its cycle-averaged dissipation at $Re = 500$ and $\bar{E}_r = 0.250$ is 1.2189×10^{-2}, about 2.5% above the 1.189×10^{-2} obtained from the time-mean profile, and its kinetic-energy fluctuation ratio is 0.04%. (The eight-phase quadrature of the temporal table gives 1.2370×10^{-2} at its native $\bar{E}_r = 0.2535$; brought to the same energy the two agree to 0.07%, so the small difference is an energy rescaling, not a quadrature error.) The temporal content of B_1 is therefore dynamically small but not negligible: it contributes a few per cent to the dissipation. The stationary transfer used in the main text is a controlled approximation, adequate for ranking the stationary controls, and Sec. S8.5 supplies the cycle-averaged value for the full control.

Norm versus peak velocity. The transfer fixes the quadratic actuation norm $E_{\text{act}} = \int_0^1 U_{\text{lid}}^2 \, dx$, not the peak wall velocity or the Reynolds number based on it. At $E^* = 0.25$ the rms wall velocity is 0.5 for every case, but the peak is $U_{\max} = 0.50$ for the sharp uniform lid, 0.52 for the tapered lid, 0.71 for the two harmonics and 0.78 for the B_1 mean profile, i.e. $Re_{\max} = U_{\max} Re = 250$, 258, 354 and 390 at $Re = 500$. For the time-dependent branches the tabulated $U_{\max}$ is that of the time-mean profile, with the maximum over the period in parentheses; it is 0.04 (1.23) for B_2 and 0.62 (1.16) for B_3. The comparisons of the main text are therefore at equal quadratic norm and fixed viscosity,

not at equal $Re_{\max}$, and the two conventions must not be mixed when the values are compared with lid-speed-based literature data.

Figure ? draws the same four transfers at both Reynolds numbers. The ordering is identical at $Re = 500$ and $Re = 1000$: the uniform lid is the most dissipative, the stationary second harmonic the least, and the mean PINN profile and the first harmonic lie between them, with reductions of 76.5% and 82.3% for the second harmonic against 73.1% and 80.3% for the PINN mean profile. The near-equality of the last two bars is the observation that motivated the stationary reduction of the B_1 branch, and the velocity-scale caveat stated in Sec. S8.3 is the reason it is not reported as a performance claim in this study.

S8.4 Grid convergence of the controlled-flow dissipation

The grid study of Main Sec. 4.3 was repeated for all four stationary controls (uniform lid, $k = 1$, $k = 2$ and the B_1 PINN mean profile) on the same three grids $N \in \{128, 256, 512\}$ at $Re = 500$, with the lid amplitude of each profile rescaled so that the actuation energy is the common value $E^* = \int_0^1 U_{\text{lid}}^2\, dx = 0.25$. The uniform lid therefore uses $U_{\text{lid}} = 0.5$, and the two harmonic controls use the amplitude $\sqrt{2E^*} = 0.7071$. All runs use the same MRT collision model, the same convergence criterion and up to 2×10^5 iterations. Because the lid amplitude differs from the $U_{\text{lid}} = 1$ uniform baseline of the main text, the dissipation values reported here are *not* comparable with those of the problem statement; they are the values that define the iso-energetic comparison.

For each control the observed order of convergence is obtained from the Richardson extrapolation

$$p_{\text{obs}} = \frac{\ln[(\varepsilon_{512} - \varepsilon_{256})/(\varepsilon_{256} - \varepsilon_{128})]}{\ln r}, \qquad r = 2, \tag{25}$$

the extrapolated value from

$$\varepsilon_\infty = \varepsilon_{512} + \frac{\varepsilon_{512} - \varepsilon_{256}}{r^{\,p_{\text{obs}}} - 1}, \tag{26}$$

and the fine-grid grid-convergence index from

$$\text{GCI}_{512} = F_s\, \frac{|(\varepsilon_{512} - \varepsilon_{256})/\varepsilon_{512}|}{r^{\,p_{\text{obs}}} - 1}, \qquad F_s = 1.25. \tag{27}$$

When the three values are not monotone, when the estimated p_{obs} falls outside $[0.1, 4]$, or when the GCI exceeds 100%, the triple is flagged and its extrapolated value is not used: in that case the smooth-error assumption underlying the estimator is not credible.

S4 Independent MRT-LBM Verification and Grid Uncertainty

S4.1 MRT-LBM Formulation

The independent lattice-Boltzmann solver uses the D2Q9 model with 9 velocity directions $\mathbf{c}_i$ ($i = 0, \ldots, 8$) defined as $\mathbf{c}_0 = (0, 0)$, $\mathbf{c}_{1-4} = (\pm 1, 0), (0, \pm 1)$, and $\mathbf{c}_{5-8} = (\pm 1, \pm 1)$ (lattice units). The lattice speed of sound is $c_s = 1/\sqrt{3} \approx 0.5774$ in lattice units. Velocities are stored in the lattice frame and divided by a reference velocity $U_{\text{ref}} = 0.05$ (lattice units) before being compared with the cavity-unit fields of the main text, so that a lid speed of unity in cavity units is 0.05 in lattice units. The corresponding reference Mach number is $\text{Ma}_{\text{ref}} = U_{\text{ref}}/c_s = 0.0866$ per unit cavity speed. The uniform lid at $E^* = 0.25$ has cavity amplitude 0.5, giving $\text{Ma}_{\text{uniform}} = 0.043$. The largest lid speed imposed anywhere in the campaign is the instantaneous B_2 value 1.23, giving $\text{Ma}_{B_2,\max} = 1.23 \times 0.0866 \approx 0.106$, within the incompressible limit ($\text{Ma} < 0.3$). The MRT collision step [4] is

$$\mathbf{m}^* = \mathbf{m} - \mathbf{S}(\mathbf{m} - \mathbf{m}^{\text{eq}}), \tag{28}$$

where the moment vector $\mathbf{m} = \mathbf{M}\mathbf{f}$, $\mathbf{M}$ is the 9×9 transformation matrix from population to moment space, and $\mathbf{S} = \text{diag}(s_0, s_1, \ldots, s_8)$. Only the two shear modes are tied to the Reynolds number, through $s_7 = s_8 = s_\nu = 2/(1 + 6\nu_{\text{latt}})$, where $\nu_{\text{latt}} = U_{\text{ref}} N/Re$ is the lattice viscosity (with N the grid size per cavity side and U_{ref} the reference lattice velocity). The conserved moments (density ρ

and momentum j_x, j_y) are exactly preserved during collision, so their relaxation rates are irrelevant ($s_0 = s_3 = s_5 = 0$ in the standard MRT formulation, or equivalently these moments are not relaxed). The remaining non-viscous, non-conserved modes keep the conventional fixed values of the D2Q9 multiple-relaxation-time model [4]: the energy mode $s_1 = 1.64$, the energy-square mode $s_2 = 1.54$, and the two energy-flux (ghost) modes $s_4 = s_6 = 1.9$. The first two fix the bulk viscosity, which therefore stays constant across the parameter sweep. These four rates are constants, independent of Re and of the lattice size, so the equation of state and the compressible corrections are unaffected by the Reynolds number; they provide numerical stability while maintaining second-order accuracy. The post-collision populations are streamed: $f_i(\mathbf{x} + \mathbf{c}_i, t + 1) = f_i^*(\mathbf{x}, t)$.

Lid boundary conditions are imposed on the node row $i = N - 1$ by half-way bounce-back with momentum injection into the two wall-facing populations ($q = 7, 8$), a velocity boundary condition of the Zou–He family [3]. The injected momentum is the commanded lid speed $U_{\mathrm{lid}}(x, t_{\mathrm{phys}})$ at that node, with the local density taken from the post-collision field. Because this scheme places the hydrodynamic wall half a lattice spacing outside the lid node row, it enforces the moving-wall condition at the wall, not the velocity at the node: the recovered streamwise velocity of row $N - 1$ differs from the commanded profile by the slip measured in Sec. S4.8 (3.2% to 11.8% of the cavity speed, resolution-dependent), and the power estimator of Sec. S4.8 is therefore evaluated on that row rather than exactly at $y = 1$. Bottom and lateral walls use standard half-way bounce-back boundary conditions.

S4.2 Ghia et al. Benchmark Validation

The MRT-LBM solver was validated against the reference benchmark data of Ghia et al. [2] for the uncontrolled square cavity at $Re = 100$ and $Re = 1000$ using a 256×256 lattice. Main Sec. 4.2 reports the L_2 and L_∞ errors in the centerline velocity profiles. The errors are within the expected accuracy of the LBM at this resolution, confirming that the solver correctly reproduces the known benchmark solutions before being applied to the controlled-flow study.

The benchmark above constrains the steady-state accuracy of the solver. Two further checks constrain its unsteady behaviour, because the branch controls are time-dependent.

Periodic steady state. The phase averages of the first and second halves of the sampling window differ, in relative L_2 norm of the streamwise velocity field, by at most 1.9×10^{-3} for B_1, 2.6×10^{-4} for B_3 and 1.5×10^{-5} for B_2 (Sec. S4.6), so each run is advanced far enough that the transient has left the sampled window, and the periodic response is reproduced independently in the two halves of that window.

Single-frequency oscillating lid. The temporal mapping itself was checked with a dedicated run in which the lid is a pure harmonic, $U_{\mathrm{lid}}(x, t) = \sin(2\pi x)\cos(\omega_1 t)$ at the same $E^* = 0.25$, $Re = 500$, $N = 256$ and $\omega_1 = \pi$ as the branch captures (`scripts/validate_stokes_analytic.py`, output `results/08_stokes_analytic/stokes_analytic.npz`). Oscillating-lid cavity flows have been simulated with the lattice Boltzmann method before [5]; here the aim is not a flow regime but a frequency-response check of the solver and of the mapping. The streamwise velocity recorded at the antinode of the lid profile then contains the forcing frequency and no other significant line: the single-sided amplitude at ω_1 is 0.927 at the lid node, i.e. the imposed unit amplitude recovered to within the 7.3% slip of the velocity boundary condition discussed in Sec. S4.8, and the wall-normal profile of that amplitude decays exponentially over the cells below the lid, with a fitted decay length of 5.4 cells and $R^2 = 0.999$ over the eleven cells nearest the wall. The half-space Stokes thickness at ω_1 is $\delta = \sqrt{2\nu/\omega_1} = 9.1$ cells. The measured layer is therefore of the right order but 41% thinner than the semi-infinite-slab value: the lid is non-uniform in x, the cavity depth is only about 28δ, and the return flow superimposes a steady shear on the oscillating component, none of which enters the second-problem idealisation. The run is used as evidence that the solver responds at the imposed frequency with a wall-attached layer of the expected order of magnitude, not as a quantitative benchmark of the layer thickness; the quantitative claims of this study rest on cycle means, which are additionally checked by the drift, quadrature and closure tests of Secs. S4.6 and S4.8.

S4.3 Control Transfer Protocol

The PINN-identified controls are transferred to the LBM solver by evaluating the control network output at the desired Re and discretizing the resulting $U_{\mathrm{lid}}(x, t_{\mathrm{phys}}, Re)$ profile onto the lattice lid nodes. For the stationary $A_2 \sin(2\pi x)$ and $\sin(\pi x)$ configurations, the lid velocity is constant in time and is applied identically at every LBM timestep.

For the B_1 branch control, two distinct profiles must be differentiated:

1. The full B_1 control $U_{\mathrm{lid}}(x, t_{\mathrm{phys}}; 500)$ used for the dedicated branch capture is rescaled to the target cycle-mean actuation energy $E^* = 0.25$. Its native, unscaled Fourier export has $\overline{E}_r = 0.2535$. It contains both stationary and time-dependent Fourier coefficients, with temporal energy fraction $f_t = 0.013$.

2. The B_1 *mean profile* $\overline{U}_{\mathrm{lid}}(x; 500)$ is obtained by temporal averaging over the normalized phase $\theta \in [0, 1]$, which removes all coefficients a_{mn} with $n \geq 1$. Specifically,

$$\overline{U}_{\mathrm{lid}}(x; 500) = \int_0^1 U_{\mathrm{lid}}(x, \theta; 500)\, d\theta = \sum_{m=1}^{N_x} a_{m0}(500) \sin(m\pi x), \tag{29}$$

   equivalently $\overline{U}_{\mathrm{lid}} = T^{-1} \int_0^T U_{\mathrm{lid}}(x, t_{\mathrm{phys}}; 500)\, dt_{\mathrm{phys}}$ in physical time. By construction, this stationary profile has $f_t = 0$. Two transfers must be distinguished. The *native* full Fourier series has actuation energy $\bar{E}_r = 0.2535$; its time-mean profile, extracted from the Fourier coefficients a_{m0}, has energy $E_{\mathrm{mean,native}} = (1 - f_t)\bar{E}_r = (1 - 0.013) \times 0.2535 = 0.2502$, which is within 0.1% of the realised energy 0.2503 of the stationary run of that profile (Table S14). When the full series is rescaled to the target $E^* = 0.25$ for the dedicated branch capture, its mean profile energy becomes $E_{\mathrm{mean}} = (1 - f_t)E^* = (1 - 0.013) \times 0.25 = 0.2468$. This small deviation arises because the PINN optimization constrains the total energy of the full control to $E^* = 0.25$, not the energy of the mean profile alone. The mean profile used independently as a stationary lid condition has energy determined solely by the a_{m0} coefficients and is 0.2503 for the native series.

The B_1 mean profile is transferred to the MRT-LBM solver and applied as a steady lid condition. No retraining or adaptation of the PINN is performed; the transfer is a one-time evaluation of the neural network. No renormalization is applied: the profile is used as computed from the Fourier coefficients.

The stationary approximation implicit in this protocol is quantified by the dedicated capture of the full control (Sec. S4.6). The B_1 branch carries a temporal energy fraction of 1.25% ($f_t = 0.013$), so the time-mean profile is not exactly the control that the PINN produces. When the full time-dependent Fourier series is transferred instead, at the same target energy $E^* = 0.25$, $Re = 500$ and $N = 256$, its cycle-mean dissipation is 1.1934×10^{-2}, i.e. 0.34% above the 1.1894×10^{-2} of the stationary run of its own time-mean profile, and the temporal fluctuations carry $R_K = 0.16\%$ of the cycle-mean kinetic energy. The two values are not the same pair of quantities, and the comparison is deliberately made between the two *cycle means*: the stationary control and the time-dependent one are then within 0.4% of each other, while the dissipation of the time-mean *field* of the unsteady run, $D(\overline{\mathbf{u}}) = 1.1719 \times 10^{-2}$, is 1.8% below its own cycle mean (Jensen gap; Sec. S4.5). The eight-bin phase quadrature of the same record reproduces the cycle mean to 0.0017%, so the agreement with the stationary profile is not a quadrature artifact. The temporal content of B_1 is therefore dynamically small: it costs well under one per cent of dissipation and does not change the ranking of Sec. S4.4.

Norm versus peak velocity. The transfer fixes the quadratic actuation norm $E_{\mathrm{act}} = \int_0^1 U_{\mathrm{lid}}^2\, dx$, not the peak wall velocity or the Reynolds number based on it. At $E^* = 0.25$ the rms wall velocity is 0.5 for every case, but the peak is $U_{\max} = 0.50$ for the sharp uniform lid, 0.52 for the tapered lid, 0.71 for the two harmonics and 0.78 for the B_1 mean profile, i.e. $Re_{\max} = U_{\max} Re = 250$, 258, 354 and 390 at $Re = 500$. For the time-dependent branches the tabulated $U_{\max}$ is that of the

Table S10: Grid convergence and iso-energetic comparison at $Re = 500$ with matched actuation energy $E^* = 0.25$ ($E_r = \int_0^1 U_{\mathrm{lid}}^2\, dx$). ε is the integrated viscous dissipation; GCI is the fine-grid Grid Convergence Index with $F_s = 1.25$. The B_1 PINN mean control is used as computed ($E_r = 0.2503$), i.e. without rescaling. A flagged GCI means that the triple does not satisfy the smooth-error assumption; this is the case for the uniform lid, discussed in the corner-singularity study.

Control	E_r	ε_{128}	ε_{256}	ε_{512}	GCI_{512} (%)	$\varepsilon_{512}/\varepsilon_{512}^{\mathrm{unif}}$
Uniform lid	0.250	0.00774	0.00948	0.01102	130.71	1.000
$k = 1$ ($\sin \pi x$)	0.250	0.00635	0.00688	0.00713	3.85	0.647
$k = 2$ ($\sin 2\pi x$)	0.250	0.01012	0.01129	0.01185	5.62	1.076
B_1 PINN mean	0.250	0.01067	0.01189	0.01250	6.01	1.134

time-mean profile, with the maximum over the period in parentheses; it is 0.04 (1.23) for B_2 and 0.62 (1.16) for B_3. The comparisons of the main text are therefore at equal quadratic norm and fixed viscosity, not at equal $Re_{\max}$, and the two conventions must not be mixed when the values are compared with lid-speed-based literature data. The same distinction applies to the fixed-lid-velocity comparison of Ref. [1], in which all profiles were normalized to a common peak wall speed rather than to a common actuation energy: its rankings are therefore not transferable to the iso-energetic comparison of Table S14 or to the stationary study of Sec. S4.4.

S4.4 Grid convergence of the controlled-flow dissipation

The grid study of Main Sec. 4.3 was repeated for all four stationary controls (uniform lid, $k = 1$, $k = 2$ and the B_1 PINN mean profile) on the same three grids $N \in \{128, 256, 512\}$ at $Re = 500$, with the lid amplitude of each profile rescaled so that the actuation energy is the common value $E^* = \int_0^1 U_{\mathrm{lid}}^2\, dx = 0.25$. The uniform lid therefore uses $U_{\mathrm{lid}} = 0.5$, and the two harmonic controls use the amplitude $\sqrt{2E^*} = 0.7071$. All runs use the same MRT collision model, the same convergence criterion and up to 2×10^5 iterations. These matched-energy results are directly comparable with the uniform lid at $U_{\mathrm{lid}} = 0.5$ used as the reference in the present study, but not with the earlier fixed-lid-velocity comparisons of Ref. [1], in which all profiles were normalized to the same peak wall speed. The dissipation values reported here are the ones that define the iso-energetic comparison.

For each control the observed order of convergence is obtained from the Richardson extrapolation

$$p_{\mathrm{obs}} = \frac{\ln[|\varepsilon_{128} - \varepsilon_{256}|/|\varepsilon_{256} - \varepsilon_{512}|]}{\ln r}, \qquad r = 2, \tag{30}$$

the extrapolated value from

$$\varepsilon_\infty = \varepsilon_{512} + \frac{\varepsilon_{512} - \varepsilon_{256}}{r^{\,p_{\mathrm{obs}}} - 1}, \tag{31}$$

and the fine-grid grid-convergence index from

$$\mathrm{GCI}_{512} = F_s\, \frac{|(\varepsilon_{512} - \varepsilon_{256})/\varepsilon_{512}|}{r^{\,p_{\mathrm{obs}}} - 1}, \qquad F_s = 1.25. \tag{32}$$

When the three values are not monotone, when the estimated p_{obs} falls outside $[0.1, 4]$, or when the GCI exceeds 100%, the triple is flagged and its extrapolated value is not used: in that case the smooth-error assumption underlying the estimator is not credible.

The energy-matched values, the observed order of convergence, the extrapolated values and the fine-grid GCIs of the four stationary controls are collected in Table S10.

The uniform lid is the weak point of this analysis, and the corner study reported in Table ? identifies the cause. Its increments between successive grids are 1.745×10^{-3} and 1.540×10^{-3}, so they decay by only 12% per refinement, giving $p_{\mathrm{obs}} = 0.18$ and $\mathrm{GCI}_{512} = 131\%$. The formal Richardson limit is 2.25×10^{-2}, more than twice the finest-grid value; we quote it only to show that it is implausible, and do not use it. The cause is the tangential-velocity discontinuity at the two upper corners, which violates the smooth-error assumption and is absent from the sine controls.

Table S11: Effect of the upper-corner velocity discontinuity on the grid convergence of the uniform lid at $Re = 500$, $E^* = 0.25$ and $N \in \{128, 256, 512\}$. The tapered lid has the same unit plateau but a raised-cosine taper $U = \frac{1}{2}[1 - \cos(\pi x/\delta)]$ of width $\delta = 0.05$ at each upper corner, so its tangential velocity is continuous and the amplitude is rescaled to the same $E_r = 0.25$. It is a different lid profile and does not replace the discontinuous lid in the iso-energetic ranking; it isolates the cause of the non-convergence.

Control	ε_{128}	ε_{256}	ε_{512}	p_{obs}	GCI_{512} (%)	$\varepsilon_{512}/\varepsilon_{512}^{\text{unif}}$
Uniform lid (discontinuous)	0.00774	0.00948	0.01102	0.18	130.71	1.000
Uniform lid, $\delta = 0.05$ taper	0.00717	0.00815	0.00866	0.96	7.79	0.786

Replacing that lid by a cosine taper of width $\delta = 0.05$ at each corner, at the same energy, restores convergence: $p_{\text{obs}} = 0.96$, $\text{GCI}_{512} = 7.8\%$, $\varepsilon_\infty = 9.20 \times 10^{-3}$. The tapered lid is a different profile from the lid specified in the problem and is not substituted for it in the ranking; it serves to isolate the singularity and to provide a converged reference.

The uniform lid is the weak point of this analysis, and the corner study reported in Table S11 identifies the cause. Its increments between successive grids are 1.745×10^{-3} and 1.540×10^{-3}, so they decay by only 12% per refinement, giving $p_{\text{obs}} = 0.18$ and $\text{GCI}_{512} = 131\%$. The formal Richardson limit is 2.25×10^{-2}, more than twice the finest-grid value; we quote it only to show that it is implausible, and do not use it. The cause is the tangential-velocity discontinuity at the two upper corners, which violates the smooth-error assumption and is absent from the sine controls. Replacing that lid by a cosine taper of width $\delta = 0.05$ at each corner, at the same energy, restores convergence: $p_{\text{obs}} = 0.96$, $\text{GCI}_{512} = 7.8\%$, $\varepsilon_\infty = 9.20 \times 10^{-3}$. The tapered lid is a different profile from the classical uniform lid and is not substituted for it in the ranking; it serves to isolate the singularity and to provide a converged reference.

The corresponding convergence of the *ratios* $\varepsilon_{\text{control}}/\varepsilon_{\text{uniform}}$, which are the quantities used in Main Sec. 5.1, is reported separately because ratio errors need not cancel. The uniform lid has a discontinuous tangential velocity at the two upper corners, so its discretization error is dominated by a local singularity that is absent from the sine controls, whose lid velocity vanishes at both corners. The errors of numerator and denominator therefore have different origins and different signs of approach, and no error cancellation can be assumed a priori. Two normalizations are therefore reported in Table ?, one by the sharp lid specified in the problem and one by the corner-regularized lid.

The corresponding convergence of the *ratios* $\varepsilon_{\text{control}}/\varepsilon_{\text{uniform}}$, which are the quantities used in Main Sec. 5.1, is reported separately because ratio errors need not cancel. The uniform lid has a discontinuous tangential velocity at the two upper corners, so its discretization error is dominated by a local singularity that is absent from the sine controls, whose lid velocity vanishes at both corners. The errors of numerator and denominator therefore have different origins and different signs of approach, and no error cancellation can be assumed a priori. Two normalizations are therefore reported in Table S12, one by the classical sharp uniform lid and one by the corner-regularized lid.

With the sharp lid as reference the three ratio triples drift by 18–21% over the triple: $k = 1$ gives $0.821 \to 0.725 \to 0.647$ with $p_{\text{obs}} = 0.28$ and a formal GCI of 72%, while the $k = 2$ and B_1 triples are monotone but converge too slowly for the estimator ($p_{\text{obs}} = 0.038$ and 0.055, outside $[0.1, 4]$) and are flagged without a GCI. With the regularized lid as reference all three converge: $0.886 \to 0.843 \to 0.823$ ($p = 1.03$, GCI 3.1%) for $k = 1$, $1.413 \to 1.384 \to 1.368$ ($p = 0.82$, GCI 1.9%) for $k = 2$ and $1.488 \to 1.458 \to 1.443$ ($p = 0.97$, GCI 1.4%) for the B_1 mean profile, with extrapolated values 0.802, 1.347 and 1.427. The sign of the ordering is the same under both references and at every grid; the magnitude is not. A single-campaign GCI built on the sharp lid would therefore misrepresent the uncertainty of the iso-energetic comparison, and two independent checks are reported in its place.

The first is an inter-campaign cross-check. The same matched-energy problem had already been solved in an earlier, independent MRT-LBM campaign, driven with a different iteration budget:

Table S12: Grid convergence of the dissipation ratio $\varepsilon/\varepsilon_{\mathrm{ref}}$ at $Re = 500$, $E^* = 0.25$, formed on each grid before the Richardson extrapolation, so the quoted GCI refers to the quantity used in the text. Two references are used: the sharp uniform lid specified in the problem, whose own dissipation grows by 16% between $N = 256$ and $N = 512$ and has no usable GCI, and a corner-regularized uniform lid (cosine taper $\delta = 0.05$, the corner table above) at the same energy. With the sharp reference all three ratios drift by 18–21% over the triple and two of them fall outside the admissible range of the estimator; with the regularized reference all three converge with $p_{\mathrm{obs}} \in [0.8, 1.1]$ and GCIs of 1.4–3.1%. A dash means the estimator is not applicable.

Reference	Control	ratio_{128}	ratio_{256}	ratio_{512}	p_{obs}	$\mathrm{GCI}_{\mathrm{ratio}}$ (%)	$\mathrm{ratio}_{\mathrm{exact}}$
sharp lid	$k = 1$ $(\sin \pi x)$	0.821	0.725	0.647	0.28	72.37	0.272
	$k = 2$ $(\sin 2\pi x)$	1.309	1.191	1.076	–	–	–
	B_1 PINN mean	1.379	1.254	1.134	–	–	–
regularized lid	$k = 1$ $(\sin \pi x)$	0.886	0.843	0.823	1.03	3.06	0.802
	$k = 2$ $(\sin 2\pi x)$	1.413	1.384	1.368	0.82	1.91	1.347
	B_1 PINN mean	1.488	1.458	1.443	0.97	1.39	1.427

that campaign reduced the optimization residual to order 10^{-10} (4.8×10^5 iterations for the uniform case) where the present runs stop at the 2×10^5 cap with residuals of 3×10^{-6} to 3×10^{-5}. Its three stationary controls at $A_2 = 0.7071$, namely the uniform, $\sin(\pi x)$ and $\sin(2\pi x)$ lid profiles, are the uniform, $k = 1$ and B_1-mean cases of the present study at the same $E^* = 0.25$, so the two data sets are directly comparable: the six stationary values agree to 0.70% or better (uniform -0.18% at $N = 256$ and -0.70% at $N = 512$; $k = 1$ -0.18%, -0.64%; $k = 2$ $+0.03\%$, -0.38%), and the two transferred branch mean fields agree to 0.01% (B_2 seed 5: 3.0977×10^{-5} against 3.0979×10^{-5}; B_3 seed 0: 5.0510×10^{-3} against 5.0514×10^{-3}). The $N = 512$ ratios agree to 0.05% ($k = 1$) and 0.3% ($k = 2$). That the dissipation is insensitive to four orders of magnitude of optimization residual shows that the reported values are set by the discretization and by the physical configuration, not by the convergence history of the solver, and the inter-campaign spread is the reproducibility error quoted in Main Sec. 5.1.

The first is an inter-run cross-check. The same matched-energy problem had already been solved in a prior MRT-LBM run with a different iteration budget: the prior run reduced the optimization residual to order 10^{-10} (4.8×10^5 iterations for the uniform case) where the present runs stop at the 2×10^5 cap with residuals of 3×10^{-6} to 3×10^{-5}. Its three stationary controls at matched energy $E^* = 0.25$ used lid amplitudes $U_{\mathrm{uniform}} = 0.5$ for the uniform profile and $A_1 = A_2 = \sqrt{2E^*} = 0.7071$ for the $\sin(\pi x)$ and $\sin(2\pi x)$ profiles, corresponding to the uniform, $k = 1$ and $k = 2$ cases of the present study at the same $E^* = 0.25$. The two data sets are directly comparable: the six stationary values agree to 0.70% or better (uniform -0.18% at $N = 256$ and -0.70% at $N = 512$; $k = 1$ -0.18%, -0.64%; $k = 2$ $+0.03\%$, -0.38%), and the two transferred branch mean fields agree to 0.01% (B_2 seed 5: 3.0977×10^{-5} against 3.0979×10^{-5}; B_3 seed 0: 5.0510×10^{-3} against 5.0514×10^{-3}). The $N = 512$ ratios agree to 0.05% ($k = 1$) and 0.3% ($k = 2$). That the dissipation is insensitive to four orders of magnitude of optimization residual shows that the reported values are set by the discretization and by the physical configuration, not by the convergence history of the solver, and the inter-campaign spread is the reproducibility error quoted in Main Sec. 5.1.

The second check is the resolution trend itself. Under the regularized reference the two penalty ratios move towards unity as the grid is refined ($k = 2$: $1.413 \to 1.384 \to 1.368$; B_1 mean: $1.488 \to 1.458 \to 1.443$), so the iso-energetic penalty carried by the second harmonic decreases with resolution instead of being an artifact of one grid, whereas the benefit ratio of the first harmonic moves away from unity ($0.886 \to 0.843 \to 0.823$), i.e. its advantage grows. The remaining drift between $N = 256$ and $N = 512$ is 2.4%, 1.2% and 1.1% for the three ratios, consistent with the quoted GCIs. Neither the penalty nor the benefit is grid-independent, and the coarse-grid magnitudes must not be read as the asymptotic ones.

S8.5 Cycle-averaged response of the temporal controls

The two temporal controls, the full Fourier series and the modified-Chebyshev series, are periodic with the physical forcing period $T = 2$. The PINN operates on the normalised phase $\theta = t_{\mathrm{phys}}/T \in [0, 1]$; the temporal basis $\cos(2n\pi\theta)$ is periodic with unit period in θ by construction, so the periodicity of the actuation is enforced analytically. The time derivative that enters the Navier–Stokes residual carries the chain-rule factor $\partial/\partial t_{\mathrm{phys}} = (1/T)\,\partial/\partial\theta$. The control is exported on $t_{\mathrm{phys}} \in [0, T]$, one full period, and advanced by 256 lattice steps. The periodicity of the resulting flow field is verified a posteriori: the phase averages of the first and second halves of the sampling window differ by at most 1.0×10^{-3} in relative field norm for all three branches (Main Fig. 7 caption). The dissipation of such a flow is not a single number: what a periodic actuation is charged for is the cycle average $\bar{\varepsilon} = T^{-1}\int_0^T \varepsilon(t_{\mathrm{phys}})\,dt_{\mathrm{phys}}$, whereas the diagnostics of Sec. S8.4 report $\varepsilon(\langle u\rangle)$, the dissipation of the *time-mean* field. The two are not interchangeable. Because ε is a convex quadratic functional of the velocity gradient, Jensen's inequality gives $\bar{\varepsilon} \geq \varepsilon(\langle u\rangle)$, with equality only in the steady limit; the gap is the dissipation carried by the fluctuations.

S4.5 Cycle-averaged response of the temporal controls

The three branch controls transferred to the lattice are periodic with the physical forcing period $T = 2$. The exported control is written on the normalized phase $\theta = t_{\mathrm{phys}}/T \in [0, 1]$; the temporal basis $\cos(2n\pi\theta)$ is periodic with unit period in θ by construction, so the periodicity of the actuation is enforced analytically. The identification itself runs on the raw training clock $\tau \in [0, 1]$ (Sec. S1.2), where the same basis reads $\cos(n\pi\tau)$ and the Navier–Stokes residual differentiates with respect to τ with no chain-rule factor; export evaluates that series at $\tau \to t_{\mathrm{phys}}$ over the complete period, so the fundamental of the exported control oscillates at $\omega_1 = 2\pi/T = \pi$.

The lattice is driven so that this physical period is the one the solver sees. The lattice spacing is $\Delta x = 1/N$ and the timestep follows from the reference velocity, $\Delta t = \Delta x/U_{\mathrm{ref}}$ with $U_{\mathrm{ref}} = 0.05$, so one physical period spans $N_T = T/\Delta t = TN/U_{\mathrm{ref}} = 40N$ lattice steps, i.e. $N_T = 10\,240$ at $N = 256$. The number of steps per period therefore grows with N in proportion to $N\Delta t$-preserving frequency, and the same $\omega_1 = \pi$ is imposed on every grid: the mapping is frequency-preserving rather than step-count-preserving. The exported lid series is sampled at 256 points over one period (`endpoint=False`) and each sample is held for $N_T/256 = 40$ consecutive lattice steps, which is the temporal resolution of the imposed signal. The run is advanced until the response is periodic, the transient being terminated by the drift test of Main Sec. 4.2 (three-period windows compared until their means change by less than 2×10^{-3}; 24 forced periods for all three branches), and 32 consecutive periods are then sampled, every eighth lattice step, into the dissipation, kinetic-energy and lid-power series and into eight equally spaced phase bins of the velocity field.

The dissipation of such a flow is not a single number: what a periodic actuation is charged for is the cycle average $\bar{\varepsilon} = T^{-1}\int_0^T \varepsilon(t_{\mathrm{phys}})\,dt_{\mathrm{phys}}$, whereas the diagnostics of Sec. S4.4 report $\varepsilon(\langle u\rangle)$, the dissipation of the *time-mean* field. The two are not interchangeable. Because ε is a convex quadratic functional of the velocity gradient, Jensen's inequality gives $\bar{\varepsilon} \geq \varepsilon(\langle u\rangle)$, with equality only in the steady limit; the gap is the dissipation carried by the fluctuations.

The cycle average is evaluated from the eight phase-averaged velocity fields stored by the solver, one every $T/8$. Because the response locks on the forcing frequency, the mean of eight equally spaced phases is an exact quadrature of the cycle average for every harmonic below the fourth, but the accuracy of the quadrature depends on how much energy the response puts in higher harmonics. This was checked directly, accumulating $\varepsilon(t)$ every fourth timestep over a window of one hundred forced periods after two hundred periods of transient. At $Re = 500$, for the quasi-steady Fourier control from the earlier eight-phase campaign (Sec. S8.4, 100-period window) the direct average is 1.236×10^{-2} against 1.237×10^{-2} from the eight phases (0.07%), and the dissipation of the mean field agrees to 0.3%; the two estimates of the ratio $\bar{\varepsilon}/\varepsilon(\langle u\rangle)$ agree to 0.2% (1.042 against 1.040). For the dedicated branch capture of Sec. S8.6 (64-period window, same solver and lattice) the corresponding value is 1.219×10^{-2}, the 1.4% difference reflecting the different sampling windows and realizations rather than a methodological discrepancy. For the strongly unsteady modified-Chebyshev control, whose fluctuations carry $R_K = 44\%$ of the kinetic energy, the direct average is 4.06×10^{-2} against

Table S13: Cycle-averaged response of the three branch controls at $Re = 500$, $N = 256$ and $E^* = 0.25$, from the captures `results/09_d2_capture/`. Each run is advanced from rest, the transient is terminated by the drift test of Main Sec. 4.2 (24 forced periods), and 32 consecutive periods are then sampled every eighth lattice step. $\bar{E}_{\mathrm{act}}$ is the realized cycle-mean actuation energy, $\overline{D}$ the direct time average of the instantaneous dissipation, A_ε the half-spread of $D(t)$ over the cycle, i.e. the variability concealed by a single cycle-mean number, $D(\overline{\mathbf{u}})$ the same functional evaluated on the time-mean field, and $\overline{D}/D(\overline{\mathbf{u}})$ their ratio. The two right-hand columns are the part of the cycle mean carried by the fluctuations and the kinetic-energy fluctuation ratio $R_K = K_{\mathrm{fluct}}/\overline{K}$.

Control	f_t	$\bar{E}_{\mathrm{act}}$	$\overline{D}$	A_ε	$D(\overline{\mathbf{u}})$	$\overline{D}/D(\overline{\mathbf{u}})$	fluct. share (%)	R_K (%)
B_1 full series	0.013	0.2500	1.19×10^{-2}	1.8×10^{-3}	1.17×10^{-2}	1.02	1.8	0.16
B_2 seed 5	0.998	0.2500	1.44×10^{-2}	1.0×10^{-2}	1.02×10^{-3}	14.18	92.9	74.88
B_3 seed 0	0.448	0.2500	1.17×10^{-2}	4.3×10^{-3}	5.41×10^{-3}	2.16	53.7	12.47

3.85×10^{-2} from the eight phases, i.e. the eight-phase value is 5.4% low: the higher harmonics that the eight phases alias are not negligible in this regime. The direct value is therefore used for $\bar{\varepsilon}$ of the Chebyshev control at $Re = 500$, and the eight-phase entries at $Re = 100$ and $Re = 1000$ are marked in Table ? as lower estimates. The half-spread A_ε quoted in the table is the amplitude of $\varepsilon(t)$ over the cycle and is the uncertainty that a single cycle-mean number conceals.

Three points follow. First, for the full Fourier control the cycle average exceeds the mean-field value by only 3% at $Re = 100$ and 4% at $Re = 500$, so the 7.4% temporal content of the B_1 branch costs a few per cent of dissipation rather than a change of regime; at $Re = 1000$ the ratio rises to 2.19, i.e. the temporal contribution becomes comparable to the mean one. Second, the modified-Chebyshev controls are not low-dissipation states. Their mean-field values in Table ? (2.4×10^{-4} to 6.0×10^{-4}) lie one to two orders of magnitude below every stationary control, but this is an artifact of time-averaging a strongly unsteady flow ($R_K = 44$–847%): the mean field retains almost no motion, while the cycle averages, 2.0×10^{-2} to 1.5×10^{-1}, are the largest of all the configurations examined. Third, the verdict on the two representations does not depend on which of the two functionals is reported: on the cycle average the Chebyshev controls are the most dissipative, not the least.

The iso-energetic ranking of Main Sec. 5.1 is therefore restricted to the stationary controls, for which a single time-independent dissipation value is well defined and for which the grid-convergence analysis of Sec. S8.4 applies. The temporal controls are reported here, separately, with their cycle mean and the amplitude of the oscillation around it.

A matched-energy ranking of the temporal branch representatives recovered by the optimization complements the stationary comparison of Sec. S8.4. The lid series of the independent-validation runs are normalized by the peak of their *time-mean* profile, a convention that fixes the amplitude of the stationary part only: the resulting cycle-mean actuation energies are 0.43 (B_1), 43.0 (B_2) and 0.42 (B_3), and the B_2 lid reaches 17 reference velocity units, outside the incompressible range of the solver. The transfers reported in Table ? follow the convention of Main Sec. 2.2 instead: a single multiplicative factor brings the cycle-mean actuation energy to $E^* = 0.25$. Because the factor is uniform in space and time, f_2, f_t and C_3, and hence the branch label, are unchanged.

Four realizations of the $Re = 500$, $E^* = 0.25$ cell were transferred to the LBM with the per-step cycle-mean protocol above: B_2 seeds 5 and 14 ($f_t = 0.998$ and 0.975) and B_3 seeds 0 and 4 ($f_t = 0.448$ and 0.637).

The cycle mean quoted in Table S13 is the direct accumulation of $\varepsilon(t)$ over the 32 sampled periods, i.e. over 327 680 lattice steps recorded at every eighth step (40 960 samples). The eight phase bins stored by the same run provide an independent estimate, and the two are compared directly: the eight-bin mean is 1.19344×10^{-2} against 1.19342×10^{-2} for B_1, 1.440818×10^{-2} against 1.440819×10^{-2} for B_2 and 1.16906×10^{-2} against 1.16901×10^{-2} for B_3, i.e. differences of $+0.0017\%$, $-8 \times 10^{-6}\%$ and $+0.0039\%$. The phase quadrature is therefore adequate for the dissipation of all three controls, which is what the harmonic budget of the next subsubsection requires: no more than

4×10^{-5} of the actuation energy of any branch sits above the fourth temporal harmonic, and eight equally spaced phases integrate harmonics one to four exactly. The lid power behaves differently, because it is a product of two quantities that both vary within a phase bin: the same eight-bin reconstruction is 0.14%, 5.5% and 3.9% below the step-by-step accumulator for B_1, B_2 and B_3. Main Table 4 therefore quotes the accumulator for $\overline{P}_{\mathrm{lid}}$ and the eight-bin value is used only as a cross-check (Sec. S4.8). The half-spread A_ε quoted in the table is the amplitude of $\varepsilon(t)$ over the cycle, computed as half the peak-to-peak excursion of the eight phase-resolved values, and is the variability that a single cycle-mean number conceals.

Three readings of the table matter. First, B_1 is quasi-stationary: its cycle mean exceeds the dissipation of its own time-mean field by 1.8% (ratio 1.02), the temporal fluctuations carry $R_K = 0.16\%$ of the cycle-mean kinetic energy, and $\varepsilon(t)$ spans a factor 1.43 across the cycle. The stationary transfer of Sec. S4.4 is therefore an adequate representation of that branch. Second, B_2 is the opposite case: 92.9% of its cycle mean is carried by the fluctuations, the ratio $\bar{\varepsilon}/\varepsilon(\langle u \rangle)$ is 14.2, the mean field retains only a quarter of the cycle-mean kinetic energy, and the instantaneous dissipation spans a factor 7.4 within one period in the step-by-step record (a factor 4.4 across the eight phase bins, each of which averages over one eighth of the period), with a half-spread $A_\varepsilon = 1.02 \times 10^{-2}$ that is ten times the dissipation of the mean field itself. Reporting such a branch by a single number is only meaningful together with that number of 7.4. Third, B_3 sits between the two: 53.7% fluctuating share, ratio 2.16, $R_K = 12.5\%$ and an in-cycle factor of 2.42. The three branches have cycle means within 2% of one another for B_1 and B_3 (1.1934×10^{-2} and 1.1690×10^{-2}) against 1.4408×10^{-2} for B_2, i.e. the most temporally complex control is the most expensive, and the two controls that keep a mean flow pay for it in the mean.

The iso-energetic ranking of Main Sec. 5.1 is therefore restricted to the stationary controls, for which a single time-independent dissipation value is well defined and for which the grid-convergence analysis of Sec. S4.4 applies. The temporal controls are reported here, separately, with their cycle mean and the amplitude of the oscillation around it. Table S14 juxtaposes the two sets at the same target energy so that they can be read against one another, not so that their grid uncertainties are equal: the stationary rows carry the GCIs of Sec. S4.4, the branch rows do not.

The branch rows follow the transfer convention of Main Sec. 2.2: the exported lid series is multiplied by a single factor that brings its cycle-mean actuation energy to $E^* = 0.25$. That factor is 0.9933 for B_1, whose native Fourier export has $\bar{E}_r = 0.2534$, and unity for B_2 and B_3, whose natives are already at 0.2500. Because the factor is uniform in space and time, f_2, f_t and C_3, and hence the branch label, are unchanged. The alternative convention of normalising a lid series by the peak of its *time-mean* profile fixes only the amplitude of the stationary part and is unusable for the oscillatory branches: for B_2, whose mean profile peaks at 0.038 while its instantaneous lid peaks at 1.23, it would multiply the signal by 26, the actuation energy by 690 and the lid speed to about 32 cavity units, far outside the incompressible range of the solver; the same convention gives 0.42 and 0.65 for B_1 and B_3. All the transfers reported here are therefore norm-matched, not peak-matched.

Each branch is represented by a single realization: the full B_1 Fourier series, B_2 seed 5 ($f_t = 0.998$) and B_3 seed 0 ($f_t = 0.448$). The identification campaign recovers several members per class (Secs. S2.3 and S3.1), and the within-class spreads reported there are the reason the rows are labelled with their seeds. They are not averaged here, and the table rows are to be read as typical members, not as class means.

The transfer is verified against the independent case study of Sec. S8.4: the mean-field dissipation of the B_2 seed-5 and B_3 seed-0 transfers reproduce the values of that earlier campaign to 0.01%, so the two independent solvers agree on the same lid series. On the cycle average, B_2 costs 3.76×10^{-2} and 3.70×10^{-2} (a within-branch spread of 1.7%), B_3 2.18×10^{-2} and 2.69×10^{-2} (a spread of 23%, i.e. the B_3 class is the less well defined of the two), and the full B_1 series 1.22×10^{-2}, i.e. the same order as its own mean field and as the stationary $k = 2$ and B_1-mean controls. The B_2 mean field, by contrast, dissipates 3.1×10^{-5} and 1.8×10^{-4}, four and two orders below its cycle average, and the cycle-mean kinetic energy of the seed-5 response is 2.47×10^{-4} for a mean field of 1.55×10^{-5}. The branch that the optimization accesses most readily at high temporal content is therefore the most dissipative one at matched energy, and the accessibility order of the branches is

Table S14: Iso-energetic comparison at $Re = 500$ and $N = 256$, all controls at a cycle-mean actuation energy $E^* = 0.25$. The four stationary rows come from the grid study of Sec. S4.4, where the response is steady and the cycle mean coincides with $D(\overline{\mathbf{u}})$ by construction. The three branch rows are the captures of Sec. S4.6. f_t is the temporal energy fraction of the imposed lid series, A_ε the half-spread of $D(t)$ over the cycle and $\overline{K}$ the cycle-mean kinetic energy. Jensen's inequality ($\overline{D} \geq D(\overline{\mathbf{u}})$, equality only in the steady limit) holds for every row, and the ratio column shows that the ordering of the totals is not the ordering of the mean fields. $\overline{K}$ is reported only for the branch captures, whose records span full cycles.

Control	f_t	$\bar{E}_{\mathrm{act}}$	$D(\overline{\mathbf{u}})$	$\overline{D}$	$\overline{D}/D(\overline{\mathbf{u}})$	A_ε	$\overline{K}$
Uniform lid	0.000	0.2500	9.48×10^{-3}	9.48×10^{-3}	1.00	0	–
$k = 1$ ($\sin \pi x$)	0.000	0.2500	6.88×10^{-3}	6.88×10^{-3}	1.00	0	–
$k = 2$ ($\sin 2\pi x$)	0.000	0.2500	1.13×10^{-2}	1.13×10^{-2}	1.00	0	–
B_1 mean profile	0.000	0.2503	1.19×10^{-2}	1.19×10^{-2}	1.00	0	–
B_1 full series	0.013	0.2500	1.17×10^{-2}	1.19×10^{-2}	1.02	1.8×10^{-3}	1.16×10^{-2}
B_2 seed 5	0.998	0.2500	1.02×10^{-3}	1.44×10^{-2}	14.18	1.0×10^{-2}	2.42×10^{-3}
B_3 seed 0	0.448	0.2500	5.41×10^{-3}	1.17×10^{-2}	2.16	4.3×10^{-3}	5.53×10^{-3}

not a performance order.

Two caveats stay open. The transfers above keep the shape of a recovered control and rescale its amplitude, so they do not show what the optimizer would find under an explicit cycle-mean energy constraint in the loss; retraining under that constraint remains future work. And they are single-grid ($N = 256$) at $Re = 500$, so the three-grid study of Sec. S8.4 does not cover them. Each transfer is averaged over one hundred forced periods after two hundred transient periods and stopped when the drift between successive windows falls below 2×10^{-3}; the four branch transfers stopped at 1.28×10^5 steps with drifts of 8.6×10^{-8} to 6.1×10^{-5}, while the quasi-steady Fourier control stopped at the largest drift of the set, 2.4×10^{-3}, which bounds the systematic error of its cycle mean. The representation comparison above uses the two PINN lid series, whose cycle-mean energies (0.25 and 0.29) are close to the target and require no rescaling. **S8.6 Field captures behind the main-text branch figures**

The three field figures of the main text (Main Figs. 6–8) and Main Table 4 are built from a dedicated capture of one representative per branch, independent of the transfers of Table ? and of the independent case study of Sec. S8.4. The three runs use the same solver, the same $N = 256$ lattice, the same $Re = 500$, the same period $T = 256$ steps and the same realised actuation energy. The lid series are the exported PINN Fourier coefficients with a single multiplicative factor, 0.9930 for B_1 ($E_r = 0.2500$), 1.0000 for B_2 seed 5 ($E_r = 0.2504$) and 1.0000 for B_3 seed 0 ($E_r = 0.2503$). Only the B_1 factor differs from unity, by 0.7%, and it rescales the effective Reynolds number of that run by the same 0.7%; no other run is rescaled, and the realised rather than the prescribed actuation energy is what the figures compare.

Each run was advanced for two hundred forced periods before a drift test on the sampled dissipation $\varepsilon(t)$, which compared two sliding windows of one hundred periods sampled every fourth step, and then for sixty-four forced periods of sampling. All three settled reaches ended at the five-hundred-period cap, with relative drifts over the last window of 1.5×10^{-3} (B_1), 2.6×10^{-9} (B_2) and 7.3×10^{-5} (B_3), after 1.28×10^5 iterations each. The longer settling, compared with the two-hundred-period protocol of the transfers above, is what resolves the B_3 drift at the 10^{-4} level. Sampling covers sixty-four periods, i.e. eight blocks of eight cycles, which are the blocks behind the uncertainties of Main Table 4.

The cycle means reproduce the earlier independent campaign: the dissipation is 1.2194×10^{-2}, 3.7616×10^{-2} and 2.1826×10^{-2} against 1.2189×10^{-2}, 3.7616×10^{-2} and 2.1824×10^{-2}, the largest difference being 0.04%, and the mean fields agree to 10^{-5}. The cycle-mean kinetic energy of the B_3 capture is 3.98×10^{-3}, 1.3% above the 3.93×10^{-3} of the earlier eight-phase quadrature; that quadrature is the less accurate estimate, since it is 5.4% low for B_2, so the new value is the one quoted here and in the main text.

Two caveats apply to every figure built on these captures. First, they are single realizations. B_2 seed 5 lies 0.021 from the centroid of its class in (f_2, f_t), against a median distance of 0.014 and a class radius of at most 0.028, and it dissipates 1.7% more than seed 14. B_3 seed 0 sits at the low-dissipation end of a class that has only two members in this cell, seed 4 dissipating 23% more. The figures therefore show a typical member of B_2 and one pole of a loosely defined B_3, never a class average, and the captions label them as single realizations. Second, the fields are captured at one grid ($N = 256$) and one Reynolds number ($Re = 500$); the three-grid study of Sec. S8.4 covers the stationary controls, not these temporal ones. The drift, the periodicity check below and the block uncertainties bound the numerical error of a cycle mean at fixed realization; they say nothing about the spread of a class, which is what the iso-energetic table reports.

S8.7 Wall-shear estimator and closure uncertainty

On the cycle average the three branches cost 1.441×10^{-2} (B_2), 1.193×10^{-2} (B_1) and 1.169×10^{-2} (B_3), i.e. 52%, 26% and 23% above the uniform lid at the same $E^* = 0.25$ and $N = 256$ (9.48×10^{-3}, Table S10), and the B_1 series lands within 0.4% of its own stationary mean profile (1.189×10^{-2}). On the mean field the ordering is not merely different but almost inverted: $D(\overline{\mathbf{u}}) = 1.02 \times 10^{-3}$ for B_2, against 1.17×10^{-2} for B_1 and 5.41×10^{-3} for B_3, and the cycle-mean kinetic energy of the B_2 response is 2.42×10^{-3} for a mean field of 6.1×10^{-4}. The small mean-field numbers of the oscillatory branches are therefore a consequence of time-averaging a strongly unsteady flow, not evidence of a low-cost state: the branch the optimization reaches most readily at high temporal content is the most expensive one on the cycle average, and the accessibility order of the branches is not a performance order.

Two caveats stay open. The transfers keep the shape of a recovered control and rescale its amplitude, so they do not show what the optimizer would find under an explicit cycle-mean energy constraint in the loss; retraining under that constraint remains future work. And they are run at $Re = 500$, on the $N = 256$ lattice and on a $N = 512$ replication of the same protocol, so the three-grid stationary study of Sec. S4.4 still does not apply to them: two grids bound a variation, they do not deliver a convergence order. That replication is reported in the next subsubsection. Each capture is advanced through the drift test of Main Sec. 4.2 (24 forced periods at a 2×10^{-3} tolerance) and averaged over 32 further periods; the relative drift of the sampled dissipation over that window is 7.5×10^{-6} (B_1), 3.3×10^{-7} (B_2) and 2.6×10^{-5} (B_3), which bounds the systematic error left by the termination of the transient at the parts-per-million level.

Grid-refinement sensitivity of the temporal captures ($N = 256 \to 512$)

The three branch controls were captured a second time on a 512×512 lattice with the protocol of Sec. S4.5 unchanged in every respect except the resolution. Because the number of lattice steps per forced period is $N_T = 40N$, the mapping stays frequency-preserving: $\omega_1 = \pi$ on both grids, with $N_T = 20\,480$ at $N = 512$, and each of the 256 exported waveform samples is held for $N_T/256 = 80$ consecutive steps instead of 40. The physical duration of a sample is therefore $T/256 = 7.8125\times10^{-3}$ on both grids, so the staircase of the imposed signal is resolution common-mode and cannot be the source of a $256 \to 512$ change. That staircase was also measured directly: an otherwise identical B_2 capture at $N = 256$, driven by the same exported waveform evaluated exactly in time instead of held for 40 steps, returns $\overline{D} = 1.4425 \times 10^{-2}$ against 1.4408×10^{-2} for the held command, i.e. +0.114%, with the fluctuating part moved by +0.12% and the kinetic energy by +0.005%. The settling window (24 periods), the sampling window (32 periods), the recording stride (every eighth step), the eight phase bins and the target norm $E^* = 0.25$ are identical; the realized cycle-mean actuation energy differs from 0.25 by less than 1×10^{-16} at both resolutions, and the viscous conversion $\nu_{\mathrm{latt}} = U_{\mathrm{ref}} N/Re$ takes care of the frequency itself, so $\nu_{\mathrm{phys}} = 1/Re = 2 \times 10^{-3}$ on both grids.

The periodic response of the finer runs is verified rather than assumed. A probe of the streamwise velocity at the cavity centre, recorded over the whole sampling window, has its spectral peak at 3.141593 for all three branches, with 97.5–99.5% of its fluctuation energy in the five spectral bins around that peak, and the dissipation signal peaks at ω_1 for the quasi-stationary B_1 and at $2\omega_1$ for

Table S15: Grid-refinement sensitivity of the temporal captures at $Re = 500$ and $E^* = 0.25$. Each branch was run twice on the same D2 protocol (Main Sec. 4.2, Sec. S4.5): the fundamental of the imposed control is $\omega_1 = \pi$ on both grids because the number of lattice steps per forced period is $N_T = 40N$, i.e. 10 240 at $N = 256$ and 20 480 at $N = 512$, and both runs use 24 settling periods and 32 sampled periods. Two grids quantify a *variation*, not a convergence order, so no extrapolation and no convergence index is computed here, and the $N = 256$ values remain the ones quoted in the main text. $\overline{D}$ is the cycle-mean dissipation, $D(\overline{\mathbf{u}})$ the same functional evaluated on the time-mean field, $\overline{D}_{\mathrm{fluct}}$ their difference, the next column the fluctuating share of the cycle mean, $\overline{K}$ the cycle-mean kinetic energy and $\overline{P}_{\mathrm{lid}}$ the lid power recovered at the boundary node. The two right-hand columns are the balance residual $100(\overline{P}_{\mathrm{lid}} - \overline{D})/\overline{D}$ and the largest discrete slip of the velocity boundary condition over the record, in per cent of the cavity speed. The third row of each block is the 256 → 512 change, in per cent or in percentage points (pt). The relative standard error of every $\overline{D}$ is below 6.9×10^{-6}, which is three to four orders of magnitude below the changes reported: the +5.4%, +7.3% and +7.1% moves of $\overline{D}$ are resolved, and so is the accompanying halving of the balance residual and the drop of the lid slip, both consistent with the first-order truncation of the one-sided wall-shear estimator (Sec. S4.8). The ordering $B_2 > B_1 > B_3$ is preserved, while the B_1–B_3 separation narrows from 2.1% to 0.5%. Generated from `results/d2_derived/resolution_sensitivity_hold.json` by `scripts/export_resolution_sensitivity_table.py`.

Control	N	$\overline{D}$	$D(\overline{\mathbf{u}})$	$\overline{D}_{\mathrm{fluct}}$	share (%)	$\overline{K}$	$\overline{P}_{\mathrm{lid}}$	residual (%)	slip (%)
B_1 **full series**	256	1.19×10^{-2}	1.17×10^{-2}	2.15×10^{-4}	1.8	1.16×10^{-2}	1.16×10^{-2}	−2.79%	6.0
	512	1.26×10^{-2}	1.23×10^{-2}	2.39×10^{-4}	1.9	1.18×10^{-2}	1.24×10^{-2}	−1.36%	3.2
change		+5.4%	+5.3%	+11.0%	+0.1 pt	+2.1%	+6.9%	+1.4 pt	−47%
B_2 **seed 5**	256	1.44×10^{-2}	1.02×10^{-3}	1.34×10^{-2}	92.9	2.42×10^{-3}	1.43×10^{-2}	−0.60%	11.5
	512	1.55×10^{-2}	1.11×10^{-3}	1.44×10^{-2}	92.8	2.53×10^{-3}	1.54×10^{-2}	−0.32%	7.2
change		+7.3%	+9.7%	+7.2%	−0.2 pt	+4.6%	+7.6%	+0.3 pt	−38%
B_3 **seed 0**	256	1.17×10^{-2}	5.41×10^{-3}	6.28×10^{-3}	53.7	5.53×10^{-3}	1.16×10^{-2}	−0.44%	11.8
	512	1.25×10^{-2}	5.65×10^{-3}	6.87×10^{-3}	54.8	5.68×10^{-3}	1.25×10^{-2}	−0.18%	6.9
change		+7.1%	+4.4%	+9.4%	+1.1 pt	+2.8%	+7.4%	+0.3 pt	−41%

B_2 and B_3, which is the quadratic response of a dissipation functional to a velocity that oscillates at ω_1. The half-window phase reproduction of the finer captures is 1.79×10^{-3}, 1.48×10^{-5} and 2.69×10^{-4} in relative L_2 norm for B_1, B_2 and B_3, against 1.87×10^{-3}, 1.46×10^{-5} and 2.61×10^{-4} at $N = 256$: refining the grid does not degrade the periodicity of the state that is averaged. Each finer run cost ≈ 13.1 h of single-threaded wall time, against ≈ 0.7 h for the corresponding $N = 256$ capture.

Table S15 is the sensitivity statement the main text uses. Every quantity is a cycle mean over the same 32 periods, and its sampling precision is far better than the changes reported: the relative standard error is below 7×10^{-6} for $\overline{D}$, $\overline{K}$ and $\overline{P}_{\mathrm{lid}}$ at both grids, and below 2×10^{-5} for $D(\overline{\mathbf{u}})$; it rises to 4×10^{-4} for the fluctuating part of B_1, which is a difference of two nearly equal large numbers there, and even that leaves the +11% change of that quantity resolved. The changes are read as resolution effects, not as sampling noise. The cycle mean of the dissipation rises by +5.40% (B_1), +7.33% (B_2) and +7.08% (B_3); the mean-field part by +5.29%, +9.70% and +4.45%; the fluctuating part by +11.02%, +7.15% and +9.36%. The three fluctuating shares move by +0.10, −0.20 and +1.06 percentage points, which is the sense in which the Jensen decomposition of Main Sec. 6.2 is grid-robust: refining the lattice changes the total cost of a branch by several per cent and its *composition* by about one. The cycle-mean kinetic energy rises by +2.1%, +4.6% and +2.8%, and the lid power by +6.9%, +7.6% and +7.4%, that is, in every case slightly faster than the dissipation, so the budget closes better on the finer grid.

Two readings of the table carry weight for the main text. First, the ranking. The ordering $B_2 > B_1 > B_3$ is preserved, and the B_1–B_3 separation, which is +2.09% at $N = 256$, shrinks to +0.48% at $N = 512$ — both changes far outside the standard error, and the second one consistent with the claim of Main Sec. 6.2 that the two branches are nearly equal in total cost even though

their temporal fractions are 0.013 and 0.448. The B_2 premium, on the other hand, is +20.7% and +23.2% over B_1 and B_3 at $N = 256$ and +22.9% and +23.5% at $N = 512$: the effect the paper is about does not move. Second, the closure. The balance residuals of Main Sec. 6.5, −2.79%, −0.60% and −0.44% at $N = 256$, become −1.36%, −0.32% and −0.18% at $N = 512$: each is divided by 2.1, 1.9 and 2.4, and the largest discrete slip of the velocity boundary condition falls from 6.0%, 11.5% and 11.8% of the cavity speed to 3.2%, 7.2% and 6.9%. A one-sided first-order wall-shear stencil and a half-way velocity boundary condition both truncate at $O(\Delta x)$, so halving Δx is expected to halve them, which is what is observed. This is direct evidence that the residual reported in Main Table 4 is an estimator error and not a physical imbalance; the estimator order is discussed in Sec. S4.8.

What the replication does *not* do is as important as what it does. Two grids do not give an observed order of convergence, so no Richardson extrapolation, no safety factor and no convergence index is computed from them, and the $N = 512$ values are not substituted for the headline numbers: Main Tables 3 and 4 and Main Figs. 6–8 remain the $N = 256$ captures, and the finer runs enter only as the quoted sensitivity band. The one place where the refinement changes the discussion is the Stokes-layer resolution argument of Sec. S4.5 and Main Sec. 6.3: at $N = 256$ the fundamental and the fourth harmonic of the imposed control fill about 9 and 4.6 cells, so the higher harmonics of B_2 and B_3 were marginally resolved; at $N = 512$ they fill 18 and 9.2, and the two branches nevertheless move by only 1.7 and 1.9 percentage points more than the quasi-stationary B_1, which has no harmonic-resolution issue at all. The 5.4% shift of B_1 is therefore the resolution floor of this solver configuration at $Re = 500$, and the extra shift attributable to the unsteady near-wall structure is at most two percentage points. The conclusion of Main Sec. 7.3 — that resolving the higher harmonics is not what sets the cost of the oscillatory branches — survives the refinement with a measured bound instead of an expected one.

Harmonic budget and second-Stokes estimate

Main Sec. 6.3 estimates the fluctuating part of the cycle-mean dissipation of the unsteady branches from the second problem of Stokes. The estimate uses the harmonic decomposition of the lid signal that is actually imposed, which is stored with each capture, so it can be reproduced here without re-running the solver. Table S16 gives the decomposition and its result; the decomposition and the estimate are produced by `scripts/stokes_scaling_estimate_d2.py` from the records of `results/09_d2_capture/`, with output `results/stokes_scaling_estimate_d2.json`. Three checks are worth stating. First, the decomposition closes: $\sum_{n\geq 0} E_n = 0.25000$ for all three branches, by construction of the transfer, and the temporal share $\sum_{n\geq 1} E_n / \sum_{n\geq 0} E_n = 0.0125$, 0.9977 and 0.4482 reproduces the classification fractions f_t of the same three controls to the last digit. The modal fractions quoted in the main text are therefore not an artifact of the PINN representation; they are recovered directly from the boundary signal that drives the lattice. The spectra are also short: beyond the fourth temporal harmonic the actuation energy is below 10^{-5} in every case (9×10^{-6} for B_1, 4×10^{-34} for B_2 and B_3), which is why the eight-bin quadrature of Sec. S4.5 is exact for the dissipation. Second, the estimate $\overline{D}_{\mathrm{St}} = \sum_{n\geq 1} E_n \sqrt{\nu\omega_n/2}$ with $\nu = 2 \times 10^{-3}$ and $\omega_n = n\pi$ gives 1.814×10^{-4}, 1.411×10^{-2} and 6.383×10^{-3}, against measured fluctuating parts $\overline{D} - D(\overline{\mathbf{u}})$ of 2.149×10^{-4}, 1.339×10^{-2} and 6.278×10^{-3}: the measured increment is 118%, 95% and 98% of the estimate, the ratio column of Table S16. For the two branches whose cost is dominated by the fluctuations the agreement is within 5% and 2%, so essentially the whole of their excess dissipation over their mean fields is predictable from the lid signal and the viscosity alone, without solving for the interior. For B_1 the two numbers are 1.8 and 2.1×10^{-4}, i.e. a 19% relative difference on an increment that is only 1.8% of its cycle mean, where the locally parallel assumption is least defensible because the response is not a layer but a modified steady recirculation. The residual mismatch between estimate and measurement reflects the streamwise gradients of the lid profile, the return flow against the side and bottom walls and the inertial transfer to the interior, none of which enters the second-problem idealisation. The estimate is a scaling bound, not a prediction, and it is quoted as such. Third, because the first harmonic of the lid dominates the temporal energy of both unsteady branches ($E_1/E^* = 0.98$ for B_2, 0.43 for

B_3), the estimate reduces to $\overline{D}_{\mathrm{fluct}} \simeq f_t E^* \sqrt{\nu\omega_1/2} = 0.01401\, f_t$, which is the collapse reported in Main Sec. 6.3; the numerical factor is $\sqrt{2\times10^{-3}\times\pi/2}\times 0.25$, and the corresponding Stokes-layer thickness is $\delta_1 = \sqrt{2\nu/\omega_1} = 0.0357$, i.e. 9.1 cells at $N = 256$, against 4.6 cells for the fourth harmonic. The layer is thus thin compared with the cavity depth, which is what makes the locally parallel assumption quantitatively useful here.

Table S16: Harmonic budget of the lid signals imposed in the three branch captures, and the second-Stokes estimate of the fluctuating part of the cycle mean (Main Sec. 6.3). $E_n = \frac{1}{2}\int_0^1 A_n^2\, dx$ is the actuation energy carried by temporal harmonic n of the imposed lid series ($n = 0$ is the stationary part), and $\overline{D}_{\mathrm{St}} = \sum_{n\geq1} E_n\sqrt{\nu\omega_n/2}$ with $\nu = 2\times10^{-3}$ and $\omega_n = n\pi$. The measured fluctuating dissipation $\overline{D} - D(\overline{\mathbf{u}})$ is the quantity the estimate is meant to bound from the actuation alone, and the ratio column expresses it as a percentage of the estimate. The last column is the Stokes-layer thickness of the fundamental and of the fourth harmonic in grid cells at $N = 256$. Produced by `scripts/stokes_scaling_estimate_d2.py` from the lid series stored in `results/09_d2_capture/`.

Branch	E_0	E_1	E_2	E_3	$\overline{D}_{\mathrm{St}}$	$\overline{D}_{\mathrm{fluct}}$	share (%)	δ_1, δ_4 (cells)
B_1 full series	0.2469	0.0029	0.0002	0.0000	1.81×10^{-4}	2.15×10^{-4}	118.5	9.1, 4.6
B_2 seed 5	0.0006	0.2443	0.0043	0.0007	1.41×10^{-2}	1.34×10^{-2}	94.9	9.1, 4.6
B_3 seed 0	0.1380	0.1081	0.0033	0.0006	6.38×10^{-3}	6.28×10^{-3}	98.3	9.1, 4.6

S4.6 Field captures behind the main-text branch figures

The three field figures of the main text (Main Figs. 6–8) and Main Table 4 are built from a dedicated capture of one representative per branch, which is also the source of the three branch rows of Table S14. The three runs use the same solver, the same $N = 256$ lattice, the same $Re = 500$, the same period $N_T = 10\,240$ lattice steps (Sec. S4.5) and the same realised actuation energy $E^* = 0.25$. The lid series are the exported PINN Fourier coefficients with a single multiplicative factor, 0.99328 for B_1 (native $\bar{E}_r = 0.25339$), 1.0000 for B_2 seed 5 (native 0.25000) and 1.0000 for B_3 seed 0 (native 0.25000). Only the B_1 factor differs from unity, by 0.7%, and it rescales the effective Reynolds number of that run by the same 0.7%; no other run is rescaled, and the realised rather than the prescribed actuation energy is what the figures compare.

Each run starts from rest and is advanced through the drift test of Main Sec. 4.2: the sampled dissipation is averaged over successive three-period windows and the transient is terminated when successive window means change by less than 2×10^{-3} in relative terms. All three branches pass at 24 forced periods (245 760 lattice steps), and 32 further periods (327 680 steps) are then sampled. The dissipation, kinetic-energy and lid-power series are recorded every eighth step (40 960 samples each) and the velocity field every eighth step into eight equally spaced phase bins (40 960 samples per bin). The relative drift of the sampled dissipation between the two halves of that window is $+1.1\times10^{-3}\%$ (B_1), $-2.8\times10^{-6}\%$ (B_2) and $+1.4\times10^{-3}\%$ (B_3); the corresponding drift of the sampled kinetic energy is $1.1\times10^{-3}\%$, $1.8\times10^{-6}\%$ and $5.6\times10^{-3}\%$. The largest slip between the imposed and the recovered lid velocity in the wall row, over the whole record, is 0.060, 0.115 and 0.118 cavity units for B_1, B_2 and B_3; it is the same boundary condition discussed in Sec. S4.8. Sampling covers thirty-two periods, i.e. four blocks of eight cycles, which are the blocks behind the uncertainties of Main Table 4: the cycle-to-cycle relative standard deviation of the dissipation is 7.0×10^{-6}, 2.4×10^{-8} and 1.2×10^{-5}, the standard deviation of the four block means is 1.2×10^{-5}, 3.7×10^{-8} and 1.9×10^{-5}, and with $t_{3,0.975} = 3.182$ the resulting 95% block interval on the cycle mean is 1.9×10^{-5}, 5.9×10^{-8} and 3.1×10^{-5} in relative terms. The three runs cost 2512, 2505 and 2537 s of wall-clock time on the hardware of Sec. S5.2.

The captured cycle means are $\overline{D} = 1.19342\times10^{-2}$, 1.44082×10^{-2} and 1.16901×10^{-2}, $\overline{K} = 1.16082\times10^{-2}$, 2.42269×10^{-3} and 5.52597×10^{-3}, and $\overline{P}_{\mathrm{lid}} = 1.16012\times10^{-2}$, 1.43222×10^{-2} and 1.16383×10^{-2} for B_1, B_2 and B_3. These are the step-by-step accumulations, and they are the values quoted in the main text: the eight-bin phase reconstruction reproduces $\overline{D}$ to within 0.004%

but returns a lid power 0.14%, 5.5% and 3.9% lower, so it is used only as a cross-check (Secs. S4.5 and S4.8).

Two figures are drawn directly from these records: Figure S1 shows the eight phase-resolved streamline fields of each capture, and Figure S2 repeats the layout of Main Fig. 6 with an independent stream-function scale in every panel.

Two caveats apply to every figure built on these captures. First, they are single realizations: the full B_1 Fourier series, B_2 seed 5 and B_3 seed 0. B_2 seed 5 lies 0.021 from the centroid of its class in (f_2, f_t), against a median distance of 0.014 and a class radius of at most 0.028, so it is a typical member of a compact class. B_3 seed 0 is a member of the residual class, which the classification rule defines by complement and whose members are heterogeneous in composition (Sec. S2.3: of the 185 campaign realizations labelled B_3, 15 are dominated by the temporal family, 8 by the stationary second harmonic and 3 by higher stationary harmonics). The figures therefore show a typical member of B_2 and one member of a loosely defined B_3, never a class average, and the captions label them as single realizations. Second, the fields shown here are captured at one grid ($N = 256$) and one Reynolds number ($Re = 500$); the cycle-mean scalars of the same three runs were replicated at $N = 512$ (Sec. S4.5), which moves $\overline{D}$ by 5.4–7.3% and preserves the ordering of the branches, but the phase-resolved fields themselves were not re-captured on the finer grid, so every figure below shows the $N = 256$ realization of each branch. The three-grid study of Sec. S4.4 covers the stationary controls, not these temporal ones. The drift, the periodicity check below and the block uncertainties bound the numerical error of a cycle mean at fixed realization; they say nothing about the spread of a class, which is what the classification statistics of Secs. S2.3 and S3.1 report.

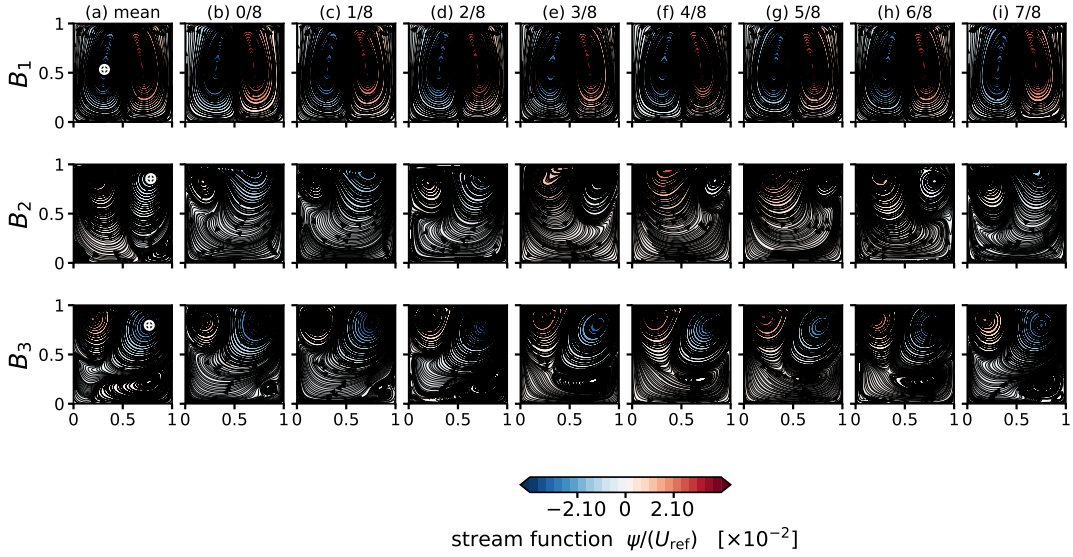


Figure S1: All eight phase bins of the three captures of Main Figs. 6–8, on the same common stream-function scale and contour levels as the main figure. Column (a) is the time mean, columns (b)–(i) the phase averages at $t/T = 0,\ 1/8, \ldots, 7/8$. The phase averages of the first and second halves of the sampling window differ, in relative L_2 norm of the streamwise velocity field, by at most 1.9×10^{-3} for B_1, 2.6×10^{-4} for B_3 and 1.5×10^{-5} for B_2, so the state is periodic and the four phases selected for the main figure are representative rather than chosen for appearance. The distance between each phase-averaged field and the time-mean field, normalized by the norm of the mean field and measured over both velocity components, spans 3.2%–10.3% for B_1, 23%–68% for B_3 and 115%–329% for B_2: B_1 is quasi-stationary throughout the cycle, whereas the instantaneous structure of B_2 is not a small perturbation of its mean. Its mean circulation is also the weakest of the three, $\max|\psi| = 0.0116$ against 0.0363 for B_1 and 0.0349 for B_3. Streamlines are tangent to the instantaneous velocity and are not particle paths.

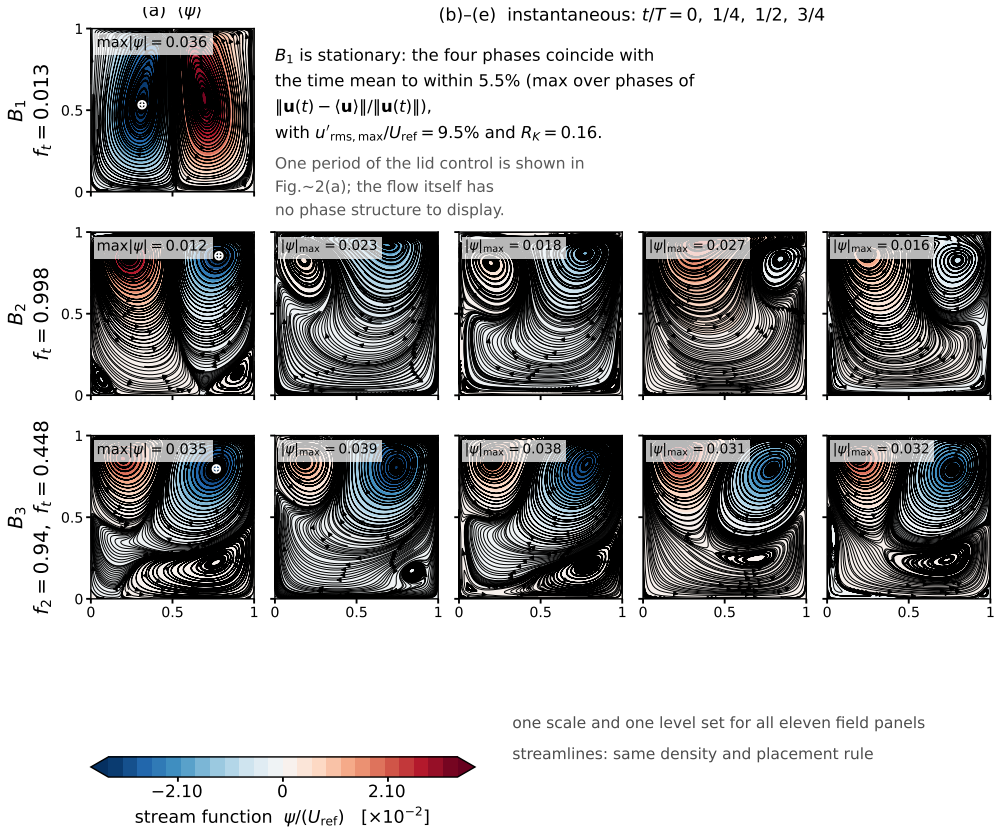


Figure S2: The layout of Main Fig. 6 with a separate stream-function scale in every panel, annotated with $\max|\psi|$ (mean panels; the instantaneous panels are annotated with their own $|\psi|_{\max}$). The common scale of the main figure is what makes the B_2 mean cell nearly invisible; this version shows its shape—a counter-rotating pair hugging the lid, with extrema at $(x, y) = (0.23, 0.85)$ and $(0.78, 0.86)$—and shows that the B_3 mean circulation sits just as high, at $(0.20, 0.86)$ and $(0.77, 0.80)$, whereas the extrema of the B_1 mean field lie at mid-depth, $y = 0.57$ and $y = 0.53$. The height at which half of the mean kinetic energy accumulates is $y = 0.89$ for both unsteady branches against $y = 0.67$ for B_1, so the two unsteady mean fields are wall-attached while the quasi-stationary one fills the cavity. The weak B_2 mean circulation ($\max|\psi| = 0.0116$, against 0.0363 and 0.0349) is the mean field the streaming diagnostic of Sec. S4.7 sets out to explain. The two versions are complementary and must be read together: only the common scale supports a quantitative comparison between branches.

S4.7 Reynolds-stress forcing of the mean flow

The main text reads the weak mean field of the oscillatory branch as consistent with a Reynolds-stress-driven steady-streaming mechanism [6]. That diagnostic rests on the forcing term alone, so the comparison is recorded here in full. The fluctuations exert $\mathbf{F} = -\nabla \cdot \langle \mathbf{u}' \otimes \mathbf{u}' \rangle$ on the mean flow, with $\mathbf{u}' = \mathbf{u}(\cdot, t) - \langle \mathbf{u} \rangle$ evaluated from the eight phase bins of the capture and the phase average weighted by the bin counts. The stresses and the divergence are computed with central differences of step $h = 1/(N - 1)$, where N denotes the number of lattice nodes per side and h is therefore the physical spacing between two consecutive nodes (the domain of length 1 is covered by $N - 1$ intervals). The three boundary cells on each side are excluded from every norm and maximum because the discrete derivative there reaches into the bounce-back walls. Table S17 reports the magnitudes.

Three numbers matter and one does not. In the lid row, $\tau_{uu} = 2.2 \times 10^{-3}$ for B_1, 2.0×10^{-1} for B_2 and 7.0×10^{-2} for B_3: the normal stress of B_2 is 92 times that of B_1 and that of B_3 is 32 times that of B_1. The forcing-to-viscous-term ratio, taken as the ratio of grid norms, is 0.259 for B_2, 0.067 for B_3 and 2.7×10^{-3} for B_1; the same ratio formed with the maxima instead of the

Table S17: Reynolds-stress forcing of the mean flow for the three branch captures at $Re = 500$, $E^* = 0.25$ and $N = 256$. τ_{uu} is the streamwise normal stress $\langle u_x'^2 \rangle$ at the lid plane, $\mathbf{F} = -\nabla \cdot \langle \mathbf{u}' \otimes \mathbf{u}' \rangle$ is the forcing that the fluctuations exert on the mean, and $\nu \nabla^2 \langle \mathbf{u} \rangle$ is the viscous term of the mean field. $\|\cdot\|$ is the root-mean-square over the fluid interior and the two ratio columns use the maximum modulus over the same region. $\max |\nabla \times \mathbf{F}|$ and its location are reported for the interior. Computed with `scripts/d2_topology_numbers.py` from the eight phase bins of `results/09_d2_capture/`; output `results/streaming_forcing_d2.json`.

Control	τ_{uu} (lid)	$\|\mathbf{F}\|$	$\|\nu \nabla^2 \langle \mathbf{u} \rangle\|$	$\|\mathbf{F}\| / \|\nu \nabla^2 \langle \mathbf{u} \rangle\|$	$\max \|\mathbf{F}\| / \max \|\nu \nabla^2 \langle \mathbf{u} \rangle\|$	$\max \|\nabla \times \mathbf{F}|$ at (x, y)
B_1 full series	2.21×10^{-3}	1.14×10^{-3}	4.29×10^{-1}	0.0027	0.004	1.1 at $(0.08, 0.99)$
B_2 seed 5	2.03×10^{-1}	5.30×10^{-2}	2.04×10^{-1}	0.2593	0.242	23.7 at $(0.36, 0.99)$
B_3 seed 0	7.00×10^{-2}	1.97×10^{-2}	2.93×10^{-1}	0.0672	0.102	-22.1 at $(0.64, 0.99)$

norms gives 0.242, 0.102 and 4.3×10^{-3}. The vorticity source of the two unsteady branches peaks just under the lid, at $x = 0.36$ for B_2 ($+23.7$) and $x = 0.64$ for B_3 (-22.1), both at $y = 0.99$, with opposite sign, while that of B_1 peaks at 1.1. What the diagnostic does *not* give is a closed mean-flow balance: the mean pressure gradient and the advection of the mean by itself were not recovered, so $\mathbf{F}$ and $\nu \nabla^2 \langle \mathbf{u} \rangle$ are compared as magnitudes, not as the two sides of an equality. The eight-bin phase average also aliases the self-product of the fourth temporal harmonic onto the mean, because that product oscillates at $8\omega_1$; the fourth harmonic holds 1.2×10^{-4} of the actuation energy of B_2 and 3.6×10^{-5} of that of B_1, so the bias is far below the factor of 47 in forcing magnitude and the factor of 97 in the forcing-to-viscous ratio that separate the branches. Computation: `scripts/d2_topology_numbers.py`; output: `results/streaming_forcing_d2.json`.

S4.8 Wall-shear estimator and closure uncertainty

The identity $\overline{P}_{\mathrm{lid}} = \overline{D}$ of the main text is exact for the continuous equations, so any residual measured between the two cycle means is a property of the discrete estimators, not of the physics. Three choices enter the estimate of P_{lid}: the velocity used at the wall node, the order of the one-sided wall-normal derivative, and the number of phases used to average the product $U_{\mathrm{lid}} \tau_w$.

Wall velocity. The transfer imposes U_{lid} through the Zou–He condition, and the velocity actually recovered at the wall node differs from the prescribed value by the discrete slip of the scheme. Using the recovered velocity $u_x(N-1, \cdot)$ in the product, rather than the prescribed U_{lid}, removes most of the residual. With a first-order one-sided derivative the cycle-mean residual $\overline{P}_{\mathrm{lid}} / \overline{D} - 1$ is -2.89%, -0.08% and -0.87% for B_1, B_2 and B_3 when the recovered velocity is used, against $+5.30\%$, $+55.40\%$ and $+43.56\%$ when the prescribed profile is used. The prescribed-velocity estimate is therefore discarded: it measures the slip of the boundary condition, not the power transmitted by the wall. All main-text values, including Main Table 4, use the recovered velocity, giving residuals of -2.9%, -0.1% and -0.9% for B_1, B_2 and B_3.

Derivative order. Replacing the first-order one-sided derivative by the three-point second-order form $(3u_{N-1} - 4u_{N-2} + u_{N-3})/(2\Delta x)$ moves the mean-field residual of B_1 from -2.87% to $+0.58\%$, i.e. it crosses zero, which shows that the truncation error of the first-order estimate is of the order of the residual itself. For the cycle means the same change gives $+1.36\%$ for B_1, but $+21.5\%$ for B_2 and $+16.6\%$ for B_3: the three-point stencil amplifies lattice-scale noise in the thin Stokes layer of the strongly oscillating lids, where successive wall-normal velocities differ by more than the signal. The second-order form is therefore better for the smooth quasi-stationary case and worse for the unsteady ones, and no single choice is optimal for all three branches. The first-order form is retained for consistency across the table, and the residual is quoted as measured.

Phase quadrature. The eight stored phase fields are sufficient for the dissipation, which is a quadratic functional of the instantaneous field, but not for the power, which is a product of two quantities that both vary within a phase bin. Averaging the recovered-velocity, first-order power over the eight phases gives -0.27%, -5.39% and -4.46% relative to the direct accumulation over 4096 steps for B_1, B_2 and B_3. The direct accumulation is therefore used for $\overline{P}_{\mathrm{lid}}$ in the main text,

and the eight-phase value is not substituted for it.

Taken together, the three checks bound the estimator uncertainty at a few per cent for the quasi-stationary branch and at roughly ten per cent for the strongly unsteady ones. This is why the main text presents $\overline{P}_{\mathrm{lid}} = \overline{D}$ as an energetic verification with an explicit estimator uncertainty, and why the conclusions rest on differences of a factor of three rather than on the third significant digit of the budget.

S9. Symbolic Regression: Configuration and LOCO Summary

The symbolic-regression campaign uses PySR (Julia backend, SymbolicRegression.jl) with the hyperparameters listed in Table ?. The full details of the search strategy, input–output specification, and per-fold expressions are given in the supplementary of Ref. ; we report here only the configuration and the leave-one-condition-out (LOCO) validation metrics that support the conclusions of the main text.

The LOCO is performed on the $E^* = 2.0$ slice, the only energy level at which all six Reynolds conditions ($Re \in \{100, 300, 500, 700, 900, 1000\}$) are populated with B_1-classified realizations. Table ? reports the per-fold metrics: the coefficient of determination R^2 and the root-mean-square error (RMSE) are evaluated on the held-out peak-normalized profile (200-point grid); the peak scale converts the expression into the dimensional mean profile; E_r is the actuation energy of that dimensional profile, evaluated for both the data and the fold expression, and their relative difference ΔE_r quantifies the amplitude error.

The mean R^2 across the six folds is 0.914, with the lowest score (0.8238) at $Re = 100$, the most demanding extrapolation case. Three folds ($Re = 300, 700, 1000$) return expressions sharing the same nested spatial argument $3.4468\,x - 1.7234$, which is antisymmetric about the cavity centre; this recurrence is the reproducible structural content of the symbolic analysis. The amplitude is not captured: ΔE_r ranges from -16.6% to $+10.5\%$, consistent with the fact that the regression target is a normalized shape. Repeating the LOCO on the $E^* = 0.5$ and $E^* = 1.0$ slices yields extrapolation failures at the corresponding edge conditions; the $E^* = 2.0$ slice is thus the only one yielding uniformly positive LOCO scores across all six Reynolds numbers, and is the one reported here.

S10. Reproducibility and Computational Pipeline

S10.1 Software Environment

Table ? lists the software environment used for all the computations reported here and in the main text.

S10.2 Hardware

PINN training was performed on a single NVIDIA A100 GPU (80 GB HBM2e) with double-precision computation enabled. The custom C++/OpenMP MRT-LBM solver was executed on a multi-core CPU workstation (8 cores, 32 GB RAM) with OpenMP thread parallelization over lattice rows.

S10.3 Computational Cost

Typical wall-clock times for the main computational components are:

- One PINN realization (300 pre-training + 3 000 joint epochs): approximately 45 minutes on a single A100 GPU.
- LBM simulation to steady state at $N = 512$: approximately 2 hours on 8 CPU cores.
- Full 185-realization broad campaign: approximately $185 \times 45 \approx 139$ GPU-hours (parallelized across 4 GPUs: approximately 35 hours wall time).
- λ_{var} campaign (90 realizations): approximately 68 GPU-hours.

S10.4 Random Seed Control

Wall velocity. The transfer imposes U_{lid} through the velocity boundary condition of Sec. S4.1: at the wall row, the two populations that would stream in from outside the domain are replaced by their bounce-back values corrected by a momentum injection proportional to the local density and to the prescribed lid velocity, which enforces that velocity at the half-way wall node. The macroscopic streamwise velocity reconstructed at that node is therefore a computed quantity, and it differs from the imposed profile by the discrete slip of the scheme; the largest slip over the whole record is 0.060, 0.115 and 0.118 cavity units for B_1, B_2 and B_3 (Sec. S4.6). Throughout this section the fields are stored as $u_x(i, j)$, where the first index is wall-normal and the second streamwise, so there are N nodes per side, the wall row is $i = N - 1$, the interior is $i \in \{1, \ldots, N - 2\}$, and the physical spacing is $h = \Delta y = \Delta x = 1/(N - 1)$; this matches the $N \times N$ array layout used by the code (e.g. $N = 256$

with the lid row at index 255). The wall shear is the first-order one-sided difference between the wall row and the fluid row below it, and the velocity in the product is the recovered wall-row velocity, so that the two factors are evaluated at the same locations as the boundary condition itself.

Two estimators of that velocity exist, the recovered one and the prescribed profile, and they bracket the balance. At fixed eight-phase quadrature and fixed first-order derivative, the cycle-mean residual $\overline{P}_{\mathrm{lid}}/\overline{D} - 1$ is -2.92%, -6.03% and -4.29% with the recovered velocity against $+3.36\%$, $+4.57\%$ and $+4.70\%$ with the prescribed profile, for B_1, B_2 and B_3. The two differ by 6%–11%, which is the size of the slip of the velocity condition, and the difference has the sign expected from it: the prescribed value over-states the boundary velocity where the scheme slips. The recovered-velocity estimator is the one accumulated by the solver and is used throughout the main text, including Main Table 4; the spread between the two is carried as the estimator uncertainty. With the direct accumulation over the sampled record the recovered-velocity residuals become -2.79%, -0.60% and -0.44%. Those three numbers carry an $O(\Delta x)$ claim, and the $N = 512$ replication of Sec. S4.5 tests that claim directly. With the identical estimator, the identical forcing and the identical sampling window, the residuals become -1.36%, -0.32% and -0.18%, i.e. divided by 2.1, 1.9 and 2.4, while the largest slip of the velocity boundary condition falls from 6.0%, 11.5% and 11.8% of the cavity speed to 3.2%, 7.2% and 6.9%. Both quantities drop at the rate their own truncation predicts, which is the evidence that the residual annotated in Main Table 4 is estimator error and not a physical imbalance. These values, together with the residual of the two recorded signals annotated in Main Fig. 8, are reproduced from the archived phase fields, mean fields and lid records of `results/09_d2_capture/capture_*_D2_N256.npz` by `scripts/d2_derived_quantities.py`, which writes `results/d2_derived/d2_summary.json`; no re-run of the solver is needed.

The discrete power estimator is therefore

$$\overline{P}_{\mathrm{lid}}^{\mathrm{rec}} = \frac{1}{T}\int_0^T \left[\sum_{j=0}^{N-1} w_j\, u_x(N-1,j)\, \tau_w(j)\, \Delta x\right] \mathrm{d}t_{\mathrm{phys}}, \tag{33}$$

with $\tau_w(j) = \nu\,[u_x(N-1,j) - u_x(N-2,j)]/\Delta y$ and trapezoidal weights $w_0 = w_{N-1} = 1/2$, $w_j = 1$ otherwise, so the two lid end nodes where the wall and the side walls meet enter with half weight. Both factors of the product are read on the lid node row $i = N-1$, and the shear is the one-sided difference between that row and the row below it. Because the half-way bounce-back that carries the boundary condition places the hydrodynamic wall half a lattice spacing outside $i = N-1$, this is the stress half a cell below the wall rather than a stress exactly at $y = 1$; the two coincide as $\Delta y \to 0$, and the discrete estimator of Main Sec. 2.2, Main Eq. (5), uses the same convention. The residuals below 3% for the direct accumulation confirm that the velocity boundary condition injects energy consistently with the discrete momentum balance, to within the discretization error of the boundary condition.

Derivative order. Replacing the first-order one-sided derivative by the three-point second-order form $\nu(3u_x(N-1,j) - 4u_x(N-2,j) + u_x(N-3,j))/(2\Delta y)$ was tested on the time-mean field, where the comparison is meaningful because the mean field by itself nearly closes the budget of the quasi-stationary branch. For B_1 it moves the mean-field residual from -3.06% to $+0.39\%$, i.e. it crosses zero, which shows that the truncation error of the first-order estimate is of the order of the residual itself. The same change applied to the mean fields of B_2 and B_3 moves their mean-field residuals from -111.2% to -112.7% and from -26.9% to -27.5%, but those two numbers are not closure errors: the mean field of an unsteady branch does not by itself balance the wall power (for B_2 the mean-field lid power is even slightly negative), which is precisely the point of the Jensen decomposition of Sec. S4.5 and of the streaming diagnostic of Sec. S4.7. The second-order form was therefore not substituted for the first-order one in the cycle-mean table, where the shear is evaluated inside a thin Stokes layer and successive wall-normal velocities differ by more than the smooth part of the signal; the first-order form is retained for consistency across the table and the residual is quoted as measured.

Phase quadrature. The eight stored phase fields provide adequate quadrature for the dissipation of all three controls: the eight-bin reconstruction of $\overline{D}$ differs from the direct accumulation over

the sampled record by $+0.0017\%$, $-8 \times 10^{-6}\%$ and $+0.0039\%$ for B_1, B_2 and B_3 (Sec. S4.5). This is a consequence of the temporal protocol: with 10 240 lattice steps per period the sampled record is dense enough that the eight bins resolve every harmonic that carries appreciable energy, the harmonics above the fourth holding less than 1×10^{-5} of the actuation energy. The direct accumulation is used nevertheless, because it costs nothing to keep. For the power estimate, which is a product of two quantities that both vary within a phase bin, the eight-bin average is measurably coarser: it returns a lid power 0.14%, 5.5% and 3.9% below the direct accumulation for B_1, B_2 and B_3. The direct accumulation is therefore used for $\overline{P}_{\mathrm{lid}}$ in the main text, and the eight-phase value is not substituted for it.

Taken together, the three checks bound the estimator uncertainty at a few per cent for the quasi-stationary branch and at about ten per cent for the strongly unsteady ones, which is the spread between the recovered-velocity and prescribed-velocity estimators; the grid refinement of Sec. S4.5 halves the recovered-velocity residual itself, which is the part the main text reports. This is why the main text presents $\overline{P}_{\mathrm{lid}} = \overline{D}$ as an energetic verification with an explicit estimator uncertainty, and why the conclusions rest on the composition of the budget—the cycle mean is 1.02, 2.16 and 14.18 times the mean-field dissipation for B_1, B_2 and B_3—rather than on the third significant digit of any single number.

S5 Reproducibility and Computational Pipeline

S5.1 Software Environment

The identification and the verification run in two separate environments: the PINN framework inherited from Ref. [1], and the independently written lattice solver of this work, which shares no deep-learning dependency with it. Table S18 lists both.

Table S18: Software environments. The upper block was used for the PINN identification (re-analysed here, not re-run), the lower block for every MRT-LBM computation reported in this article and in the main text, including the three branch captures of Sec. S4.6.

Component	Version
Identification (PINN)	
Python	3.10.12
PyTorch	2.0.1+cu118
NumPy	1.24.3
SciPy	1.11.1
Verification (MRT-LBM)	
Python	3.14.5
NumPy	2.5.3
SciPy	1.18.1
Numba (MRT-LBM kernels)	0.68.1, LLVM JIT, `parallel=True`, float64
Matplotlib (figures)	3.11.2, `Agg` backend
Solver	custom D2Q9 MRT, half-way bounce-back with lid momentum injection

S5.2 Hardware

PINN training was performed on a single NVIDIA A100 GPU (80 GB HBM2e) with double-precision computation enabled. The MRT-LBM solver is a NumPy/Numba implementation whose collision and streaming kernels are just-in-time compiled in double precision by the LLVM backend with `parallel=True`; it is a CPU code, no GPU is used for it. The populations of the nine velocities are stored as a single $9 \times N \times N$ array and the parallelisation is over lattice sites, which the Numba runtime distributes over the available cores; no hand-written threading is used. The three branch captures of Sec. S4.6 ran as single-process jobs on a 16-thread x64 desktop processor with 16 GB of RAM, and the wall-clock times quoted there are from that backend.

S5.3 Computational Cost

Typical wall-clock times for the main computational components are:

- One PINN realization (300 pre-training + 3 000 joint epochs): approximately 45 minutes on a single A100 GPU.
- Full 185-realization broad campaign: approximately $185 \times 45 \approx 139$ GPU-hours (parallelized across 4 GPUs: approximately 35 hours wall time).
- λ_{var} campaign (90 realizations): approximately 68 GPU-hours.
- One stationary MRT-LBM run to steady state at $N = 512$: approximately 2 hours on the CPU backend.
- One branch capture at $N = 256$ under the temporal protocol (245 760 warm-up steps plus 327 680 sampled steps, i.e. 573 440 lattice steps): 2512, 2505 and 2537 s for B_1, B_2 and B_3, of which about 916 s are the warm-up. The three runs together therefore cost a little over two CPU-hours, which is the reason the temporal comparison is reported for one realization per branch rather than for a seed ensemble.

- One branch capture at $N = 512$ under the same protocol (56 forced periods at 20 480 lattice steps, i.e. 1 146 880 lattice steps): 47 125, 47 144 and 47 051 s for B_1, B_2 and B_3, about 13.1 h each and 39 CPU-hours for the replication, which is why the refined campaign covers the three branch representatives and the resolution sensitivity is quoted as a band rather than extrapolated.

S5.4 Random Seed Control

Each realization is assigned a unique integer seed $s \in \{0, 1, \ldots, N_s - 1\}$. At the beginning of each realization the seed is passed to the pseudorandom generators of the deep-learning framework, of the array library and of the standard random module, and, for multi-GPU runs, to the device generators as well. Each realization is therefore strictly reproducible from its seed number.

S10.5 Inventory of Numerical Experiments

S5.5 Inventory of Numerical Experiments

Table S19 lists every numerical campaign reported in the main text and in this document, with the method used and the number of runs. The purpose is to make the campaign structure auditable: the run counts are stated per study and are not summed, because the branch-capture representatives reappear among the iso-energetic transfers, and because the capacity, truncation and aspect-ratio studies re-run the reference protocol with a single factor changed.

The two zero-run rows are re-analyses of runs already listed: the classification sensitivity relabels the 185 broad-campaign outcomes, and the ratio GCI is formed from the dissipation values of the iso-energetic campaign. The stationary iso-energetic row counts four controls on three grids; the corner-taper row adds a fifth profile on the same three grids. The LOCO row counts symbolic fits, not simulations.

S10.6 Data Availability

The cell-aggregated record of all 185 runs—realization counts, branch frequencies, modal statistics and energy calibration per cell—is available in the GitHub repository (`https://github.com/Ahmedalgeria778/PoF_cavity_dissipation_study`); the seed-level records retained for the multiplicity and ablation studies are reported in Tables ? and ?. Further details are available from the authors upon reasonable request.

The five zero-run rows are re-analyses of runs already listed: the classification sensitivity relabels the 185 broad-campaign outcomes, the ratio GCI is formed from the dissipation values of the iso-energetic campaign, the second-Stokes scaling decomposes the lid series stored with the branch captures, the lid spatial decomposition reads the same series in the spatial sine basis, and the steady-streaming forcing uses only the eight phase bins of the three captures. The stationary iso-energetic row counts four controls on three grids; the corner-taper row adds a fifth profile on the same three grids. The cycle-mean table of Sec. S4.5 (Table S13) and the branch rows of Table S14 are both read from the three branch captures, so they add no runs of their own; the stationary rows of the latter are taken from the iso-energetic campaign at $N = 256$. The three $N = 512$ captures re-run the same three controls on the finer lattice under an unchanged temporal protocol, and the exact-in-time control re-runs B_2 at $N = 256$ with only the sampling of the imposed command changed; neither adds a realization, and neither replaces a headline number.

S5.6 Data Availability

The three exported lid waveforms used by the temporal campaign are archived in this repository as `data/lid_series/`, with a manifest giving their SHA-256 checksums, array shapes, the scaling applied before transfer and the realised cycle-mean actuation energy (Sec. S4.6). The three captures themselves—the step-by-step dissipation, kinetic-energy and lid-power series, the imposed and recovered lid records, the eight phase-resolved fields and the time-mean and r.m.s. fields—are the archives `results/09_d2_capture/capture_*_D2_N256.npz`; their run logs (`results/09_d2_capture/capture_d2_.log`), the smoke test `results/07_d2_smoke/` and the single-frequency verification `results/08_stokes_analytic/` are committed, whereas the six capture archives total about 0.4 GB, exceed the per-file limit of the git tree and are supplied

Table S19: Inventory of the numerical experiments reported in this article. Runs are counted per campaign. Several studies re-use the same trained control, the same seeds or the same lattice fields, so the column is not additive and no deduplicated grand total is claimed.

Campaign	Method	Runs	Reported in
Broad (Re, E^*) campaign	PINN	185	Main Sec. 3.2; GitHub repository
Variance-regularization sweep	PINN	90	Ref. [1]; GitHub repository
Ten-seed multiplicity	PINN	10	Table S4
Network capacity	PINN	3	Table S7
Control truncation	PINN	6	Table S8
Cavity aspect ratio	PINN	3	Table S9
Fourier vs. modified Chebyshev	PINN	6	Ref. [1]
Stationary iso-energetic	LBM-MRT	12	Main Sec. 5.1; Table S10
Corner-taper control	LBM-MRT	3	Table S11
Branch captures (temporal protocol)	LBM-MRT	3	Main Secs. 6.2–6.5; Main Table 4; Tables S13 and S14
Branch captures at $N = 512$	LBM-MRT	3	Sec. S4.5; Table S15
Exact-in-time command control (B_2, $N = 256$)	LBM-MRT	1	Sec. S4.5
Single-frequency lid test	LBM-MRT	1	Sec. S4.2
Branch iso-energetic transfers	LBM-MRT	4	Table S14
Classification sensitivity	reanalysis	0	Table S2
Dissipation-ratio GCI	derived	0	Table S12
Second-Stokes scaling	derived	0	Table S16
Lid spatial decomposition	derived	0	Table S3
Steady-streaming forcing	derived	0	Table S17

as files alongside the repository and from the authors on reasonable request. Every derived quantity quoted in the main text and in this supplement is written to `results/d2_derived/` (`d2_summary.json`, `d2_extra.json`, `topology_numbers.txt`), `results/streaming_forcing_d2.json` and `results/stokes_scaling_estimate_d2.json` by the scripts named in Secs. S4.5–S4.8, and those records, which are what the tables and figures are generated from, are committed. The $N = 512$ replication adds four kinds of artifact to the same locations: the three captures `capture_*_D2_N512.npz`, the resampled command of Sec. S4.5 with its build report in `data/lid_series_n512/` (committed), the exact-in-time control `capture_B2_D2_N256_exact.npz`, and the two sensitivity records `resolution_sensitivity_hold.json` and `staircase_control_B2_N256.json` of `results/d2_derived/` (committed). Those are written by `build_lid_series_n512.py`, `run_branch_capture_d2.py`, `compare_n256_n512_d2.py`, `compare_forcing_staircase.py` and `export_resolution_sensitivity_table.py`, all in `scripts/`; the third of them runs the acceptance gate quoted in Sec. S4.5 and the last writes Table S15. The stationary and grid-convergence records are in `results/p1_gci_iso/`, the single-frequency lid test of Sec. S4.2 in `results/08_stokes_analytic/`, and the appendix tables regenerated from these artifacts in `supplementary/data_appendix/`. The cell-aggregated record of all 185 identification runs—realization counts, branch frequencies, modal statistics and energy calibration per cell—is available in the same GitHub repository (`https://github.com/Ahmedalgeria778/PoF_cavity_dissipation_study`); the seed-level records retained for the multiplicity and ablation studies are reported in Tables S4 and S5. Further details are available from the authors upon reasonable request.

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Only the works cited above are listed; the full bibliography of the companion manuscript is not repeated here.

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